

Towards Compact UOV-Based MQ Signatures: Rectangular and Lifted Whipping Structures

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Abstract

Multivariate quadratic (MQ) signatures offer fast signing and verification with short signatures, but their practicality is often limited by large public keys. Recent schemes, such as MAYO, address this limitation by employing the "whipping" technique. This method utilizes emulsifier matrices as the core component underlying Beullens' MAYO scheme to expand a mini-UOV map into a larger one while ensuring that signing reduces to solving a linear system that is full-rank with high probability. In this work, we focus on modifying these fundamental emulsifier matrices themselves to achieve better performance and smaller key sizes. First, we propose lifting the emulsifier matrices to an extension field while maintaining the base UOV map over the ground field. By leveraging the whipping technique to keep the variable-to-equation ratio close to one, this structural modification effectively avoids known lifted system attacks. Second, we enhance rectangular emulsifier matrices originally introduced in prior work with a structured block design that accelerates signing and verification while preserving the necessary full-rank behavior. This approach allows the underlying UOV instance to utilize fewer equations, yielding significantly smaller public keys and potentially faster operations. By combining both techniques, we design a new variant MAYO_L and provide a detailed security analysis against known forgery and key-recovery attacks, and propose parameter sets that improve public key and signature sizes at comparable security levels. Finally, we discuss the applicability of our lifting improvement to SNOVA, demonstrating that this enhancement can be integrated into other UOV-based schemes employing the whipping technique.

Keywords: Multivariate cryptography, UOV signatures, post-quantum cryptography, whipping technique.

1 Introduction

Public-key cryptography forms the backbone of modern secure communication, enabling key exchange, digital signatures, and authentication over the Internet. Widely deployed methods such as RSA and elliptic curve cryptography (ECC) rely on the assumed hardness of integer factorization and the discrete logarithm problem. However, Shor's quantum algorithm [1] demonstrates that these problems can be solved in polynomial time on a sufficiently large-scale quantum computer, posing a serious threat to the security of current public-key infrastructures.

Although large-scale fault-tolerant quantum computers are not yet available, encrypted data intercepted today could be stored and decrypted in the future when quantum capabilities mature—a threat often referred to as harvest now, decrypt later. This motivates the development of

post-quantum cryptography (PQC), which aims to design classical cryptographic schemes secure against both classical and quantum adversaries. In response to this need, NIST has initiated a standardization competition [2] and an additional round for digital signatures [3].

Multivariate Public-Key Cryptography (MPKC), based on the difficulty of solving systems of quadratic multivariate equations over finite fields, is considered a promising candidate resistant to quantum computers. The hardness of the MQ problem has been shown to be NP-complete in [4].

The Oil and Vinegar (OV) signature scheme, proposed by Patarin in 1997 [5], is a digital signature scheme based on the MQ problem. OV was inspired by the Linearization Equation Attack (LEA) [6] on the C* cryptosystem [7]. However, the scheme was broken by Kipnis and Shamir in [8], leading to the unbalanced version (UOV) [9]. A notable feature of the UOV scheme is that it has undergone over 20 years of cryptanalysis and remains secure. Additionally, the scheme offers very fast signing and verification, although it has the drawback of a large public key size.

To overcome this drawback, several modifications have been applied to UOV:

- Petzoldt’s key generation technique [10] allows generating part of the public key from a random seed.
- Beullens’ lifting technique [11] performs key generation over a small field but carries out signing and verification over an extension field. This technique led to the LUOV scheme, which was broken by NSDA in [12].
- Using structured block matrices to reduce key sizes: BAC-UOV [13], QR-UOV [14].
- Using the whipping technique to obtain a smaller oil space, thereby reducing key size: MAYO [15], SNOVA [16].

Our Contribution: In this paper, we propose two improvements applicable to the whipping technique introduced by Beullens [15]. Crucially, both of our contributions focus on modifying the emulsifier matrices themselves the fundamental building block of MAYO scheme to achieve better performance, smaller key and signature sizes.

- First, a lifting improvement for the emulsifier maps. Although LUOV was broken by NSDA, we observe that the attack requires the ratio between the number of variables and equations to be ≥ 2 . By employing the whipping technique, which maintains the variable-to-equation ratio close to one, this condition is naturally avoided, rendering lifted-system attacks ineffective. Thus, we demonstrate that lifting can be safely resurrected within the whipped UOV framework, providing the key size benefits of lifting without compromising security. Furthermore, we examine the impact of this lifting enhancement on existing emulsifier matrix modifications—specifically, the rectangular construction [17, 18] and the zero-matrix approach [19]. Our analysis shows that lifting strengthens resistance against forgery attacks, enabling safer adoption of more aggressive parameter choices in these modifications and thereby improving their overall effectiveness.
- Second, we propose an enhancement for rectangular emulsifier matrices. The rectangular construction itself was previously introduced in [17, 18]; we build upon this idea by incorporating a structured block design (including zero matrices, scalar block matrices, and block-circulant matrices) that significantly accelerates signing and verification while maintaining the full-rank property of the resulting map.

By combining both improvements—lifting the emulsifier matrices to an extension field and enhancing rectangular emulsifier matrices with structured design—we obtain a new signature scheme,

which we call MAYO_L^- . We provide a detailed security analysis of MAYO_L^- against all known attacks, including direct attacks, collision attacks, and key-recovery attacks (KipnisShamir, Reconciliation, Intersection, Rectangular MinRank, and Wedge attacks). Our parameter sets are chosen to achieve the same security level as the original MAYO while offering significantly smaller key and signature sizes. Furthermore, we prove the security of the scheme by reducing it to the assumed hardness of the UOV problem and the newly introduced MT-WMQ $_L^-$ problem.

Finally, we discuss the applicability of our lifting improvement to the SNOVA signature scheme, demonstrating that this enhancement can be integrated seamlessly into other UOV-based schemes employing the whipping technique.

Organization of the paper. The remainder of this paper is organized as follows. Section 2 recalls necessary preliminaries, including notation, the MQ problem, and relevant algorithms. Section 3 describes the UOV and MAYO signature schemes, with emphasis on the whipping technique and emulsifier matrices. Section 4 provides a comprehensive security analysis of MAYO, covering both forgery and key-recovery attacks. Section 5 reviews existing algorithms for solving lifted multivariate quadratic systems, including SDA, NSDA, and NSDA+. Section 6 presents our two enhancements to the whipping technique: the lifting improvement for emulsifier matrices in Section 6.1, and the structured enhancement for rectangular emulsifier matrices in Section 6.2. Section 7 introduces our new variant MAYO_L^- including its description, security proof, security analysis against known attacks, and parameter choices. Section 8 discusses the applicability of our lifting improvement to the SNOVA signature scheme. Finally, Section 9 concludes the paper.

2 Preliminaries

Notation. We denote by $\mathbb{F}_q$ the finite field with q elements. Matrices and vectors are written in boldface font: $\mathbf{M}$ and $\mathbf{v}$, respectively. Additionally, vectors are column vectors by default. Furthermore, the $n \times n$ identity matrix is denoted by $\mathbf{I}_n$. $\Lambda_{\mathbf{M}}^{(n)}$ denotes the block-diagonal matrix with n copies of $\mathbf{M}$ along the diagonal. For a vector space $V \subset \mathbb{F}_q^n$, we define its orthogonal complement $V^\perp$ as the space containing all vectors orthogonal to every vector in V , i.e., $V^\perp = \{a \mid \langle a, b \rangle = 0, \forall b \in V\}$.

Rectangular Transformation. Let $(\mathbf{M}_1, \dots, \mathbf{M}_m)^\top$ be a set of m matrices of size $n \times n$. These matrices $\mathbf{M}_i$ can be decomposed column-wise as

$$\mathbf{M}_i := (\mathbf{M}_i^{\text{col}(1)}, \dots, \mathbf{M}_i^{\text{col}(n)})$$

where $\mathbf{M}_i^{\text{col}(j)}$ is the j -th column of $\mathbf{M}_i$. We define the transformation Rect_{col} on $(\mathbf{M}_1, \dots, \mathbf{M}_m)$ as

$$\text{Rect}_{\text{col}}(\mathbf{M}_1, \dots, \mathbf{M}_m) := ((\mathbf{M}_1^{\text{col}(1)}, \dots, \mathbf{M}_m^{\text{col}(1)}), \dots, (\mathbf{M}_1^{\text{col}(n)}, \dots, \mathbf{M}_m^{\text{col}(n)})).$$

Multivariate quadratic maps. A multivariate quadratic map $\mathcal{P} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ is a system of m multivariate quadratic equations $p_1(\mathbf{x}), \dots, p_m(\mathbf{x})$ where each equation consists of n variables $\mathbf{x} = (x_1, \dots, x_n)$ and coefficients are in the finite field $\mathbb{F}_q$. We denote $\mathbf{MQ}_{n,m,q}$ as the set of multivariate quadratic maps consisting of m equations in n variables over $\mathbb{F}_q$.

MQ problem. Let $\mathcal{P} \in \mathbf{MQ}_{n,m,q}$ be a multivariate quadratic map and let $\mathbf{t} \in \mathbb{F}_q^m$ be a target vector. The MQ problem asks to find a solution $\mathbf{s}$ such that

$$\mathcal{P}(\mathbf{s}) = \mathbf{t}.$$

For convenience, we denote by $MQ(n, m, q)$ the computational complexity of solving an MQ instance consisting of m quadratic equations in n variables over $\mathbb{F}_q$.

For the MQ problem with $n \leq m$, it can be solved using algorithms such as XL [20] or F4 [21], F5 [22]. A prominent tool recently used to estimate the cost is the Hybrid Block Wiedemann XL algorithm [23]. The cost of this algorithm is

$$\min_k q^k \cdot 3 \cdot \binom{n-k+2}{2} \cdot \binom{D_{n-k,m} + n - k}{D_{n-k,m}} (2 \log_2(q)^2 + \log_2(q)),$$

where k is the number of variables randomly guessed to minimize the cost, and $D_{n-k,m}$ is the operating degree of the XL algorithm, which can be computed as the smallest integer d such that the coefficient of t^d in the series

$$\frac{(1-t^2)^m}{(1-t)^{n-k+1}}$$

is non-positive. By applying Grover's algorithm [24], the number of guessed variables can be reduced by half, and therefore the quantum complexity is estimated as

$$\min_k q^{k/2} \cdot 3 \cdot \binom{n-k+2}{2} \cdot \binom{D_{n-k,m} + n - k}{D_{n-k,m}} (2 \log_2(q)^2 + \log_2(q)).$$

For the MQ problem with $n > m$, since the number of solutions is about q^{n-m} but only one solution is required, one can randomly guess $n - m$ variables and reduce the problem to solving an MQ system with m variables and m equations over $\mathbb{F}_q$. However, when the ratio n/m is sufficiently large, algorithms such as TW [25] or the currently best known algorithm by Hashimoto [26] can be used to reduce the problem to solving easier MQ instances.

More specifically, let α and l be positive integers satisfying $n > (\alpha + 1)(m - l - \alpha + 1)$ and $n > \alpha(m - l) - (\alpha - 1)^2 + l$. Then the cost of applying the Hashimoto algorithm is

$$q^l \cdot [MQ(m - \alpha - l, m - \alpha, q) + MQ(\alpha - 1, \alpha - 1, q)] + (m - \alpha - l + 1) \cdot MQ(\alpha, \alpha, q).$$

Here, Grover's algorithm can also be applied to reduce the complexity to

$$q^{l/2} \cdot [MQ(m - \alpha - l, m - \alpha, q) + MQ(\alpha - 1, \alpha - 1, q)] + (m - \alpha - l + 1) \cdot MQ(\alpha, \alpha, q).$$

Polar forms. For a multivariate quadratic polynomial $p(\mathbf{x})$, we define its polar form as

$$p'(\mathbf{x}, \mathbf{y}) := p(\mathbf{x} + \mathbf{y}) - p(\mathbf{x}) - p(\mathbf{y}) + p(0).$$

Similarly, we define the polar form of a multivariate quadratic map $\mathcal{P}(\mathbf{x}) = p_1(\mathbf{x}), \dots, p_m(\mathbf{x})$ as

$$\mathcal{P}'(\mathbf{x}, \mathbf{y}) = p'_1(\mathbf{x}, \mathbf{y}), \dots, p'_m(\mathbf{x}, \mathbf{y}).$$

Macaulay matrix. Let $N = \frac{n(n+1)}{2}$ denote the number of distinct monomials of degree 2 in n variables $x_1, \dots, x_n$, ordered according to a fixed convention (e.g., lexicographic order). For a homogeneous quadratic polynomial $p(\mathbf{x}) = \sum_{1 \leq i \leq j \leq n} c_{i,j} x_i x_j$, its coefficient vector $c_p \in \mathbb{F}_q^N$ is the row vector whose entries are the coefficients $c_{i,j}$ arranged according to the prescribed monomial order. For a system $\mathcal{P} = (p_1, \dots, p_m)$ of m homogeneous quadratic polynomials, the Macaulay matrix $\mathbf{M}_{\mathcal{P}} \in \mathbb{F}_q^{m \times N}$ is the matrix whose ℓ -th row is the coefficient vector of p_ℓ .

$\mathcal{H}$ -Truncation. Let a be an element of $\mathbb{F}[t]/g(t)$ with the fixed representation

$$a = \sum_{i=0}^l a_i t^i.$$

Given an index set $\mathcal{H}$, we define its $\mathcal{H}$ -truncation as

$$\bar{a}^{\mathcal{H}} = \sum_{i \in \mathcal{H} \cap \{0, \dots, l\}} a_i t^i.$$

Similarly, we define the $\mathcal{H}$ -truncation of a quadratic polynomial $p(\mathbf{x})$ as

$$\bar{p}^{\mathcal{H}}(\mathbf{x}) = \sum_{i=1}^n \sum_{j=i}^n \bar{a}_{i,j}^{\mathcal{H}} x_i x_j + \sum_{i=1}^n \bar{b}_i^{\mathcal{H}} x_i + \bar{c}^{\mathcal{H}}.$$

Finally, for a polynomial system $\mathcal{P}$, we define the $\mathcal{H}$ -truncation of the entire system as

$$\bar{\mathcal{P}}^{\mathcal{H}}(\mathbf{x}) = (\bar{p}_1^{\mathcal{H}}(\mathbf{x}), \bar{p}_2^{\mathcal{H}}(\mathbf{x}), \dots, \bar{p}_m^{\mathcal{H}}(\mathbf{x})).$$

When the set $\mathcal{H}$ contains only a single element $\{c\}$, we simply write $\bar{a}^c$ instead of $\bar{a}^{\{c\}}$.

Lemma 1 ([27]). *Let A and B be two finite sets and $\mathcal{Q} : A \rightarrow B$ be a random map. For each $b \in B$, the probability that $\mathcal{Q}^{-1}(b)$ is non-empty is approximately $1 - e^{-|A|/|B|}$.*

3 UOV and MAYO Signature Schemes

3.1 The UOV Signature Scheme

3.1.1 Traditional description of UOV

Let n and o be integers such that $n > o > 0$. We define $\mathbf{x} = (\mathbf{x}_v, \mathbf{x}_o)$ where $\mathbf{x}_v = (x_1, \dots, x_{n-o})$ are called the vinegar variables and $\mathbf{x}_o = (x_{n-o+1}, \dots, x_n)$ are called the oil variables.

The secret key of UOV is a pair $(\mathcal{F}, \mathcal{T})$ where $\mathcal{T}$ is an invertible linear map and $\mathcal{F} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ is the central map represented by m equations $f_1, \dots, f_m$ as follows:

$$f_l(\mathbf{x}) = \sum_{i=1}^n \sum_{j=i}^{n-o} \alpha_{i,j}^{(l)} x_i x_j \quad (1 \leq l \leq m).$$

It is easy to see that this central map can be inverted efficiently by randomly fixing the $n - o$ vinegar variables $\mathbf{x}_v$. If $o \geq m$, the resulting linear system admits a solution.

The public key corresponding to $(\mathcal{F}, \mathcal{T})$ is the multivariate quadratic map $\mathcal{P} = \mathcal{F} \circ \mathcal{T}$. We denote by $\mathbf{P}_1, \dots, \mathbf{P}_m$ and $\mathbf{F}_1, \dots, \mathbf{F}_m$ the matrices representing the quadratic parts of the public and secret systems, respectively. In addition, let $\mathbf{T}$ be the matrix representing the linear map $\mathcal{T}$. Therefore, the public map can be computed as

$$\mathbf{P}_i = \mathbf{T}^\top \mathbf{F}_i \mathbf{T} \quad (1 \leq i \leq m).$$

To sign a message M , we first find $\mathbf{y}$ such that $\mathcal{F}(\mathbf{y}) = H(M \| \text{salt})$ where H is a cryptographic hash function and salt is a randomly chosen value for each signature. The final signature is $\mathbf{s} = \mathbf{T}^{-1}(\mathbf{y})$.

To verify a signature, we compute $\mathbf{t} = H(M \| \text{salt})$ and $\mathbf{t}' = \mathcal{P}(\mathbf{s})$. If $\mathbf{t} = \mathbf{t}'$, the signature is accepted; otherwise, it is rejected.

3.1.2 Modern description of UOV

For convenience in the security analysis as well as in describing the whipping technique, this section recalls the description of UOV from Beullens [28]. The trapdoor function of UOV is a multivariate quadratic map $\mathcal{P} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ that vanishes on a linear subspace $\mathcal{O} \subset \mathbb{F}_q^n$ of dimension $\dim(\mathcal{O}) = o = m$, i.e.,

$$\mathcal{P}(\mathbf{o}) = 0 \quad \text{for all } \mathbf{o} \in \mathcal{O}.$$

Given a target $\mathbf{t} = H(M \| \text{salt}) \in \mathbb{F}_q^m$, we find a preimage of $\mathcal{P}(\mathbf{x}) = \mathbf{t}$ by choosing a random vector $\mathbf{v} \in \mathbb{F}_q^n$ and solving the system $\mathcal{P}(\mathbf{v} + \mathbf{o}) = \mathbf{t}$ to obtain a vector $\mathbf{o} \in \mathcal{O}$. This can be done by solving a linear system in $\mathbf{o}$ since

$$\mathcal{P}(\mathbf{v} + \mathbf{o}) = \mathcal{P}(\mathbf{v}) + \mathcal{P}(\mathbf{o}) + \mathcal{P}'(\mathbf{v}, \mathbf{o}) = \mathbf{t}.$$

With high probability, this system has a solution; otherwise, we sample another vector $\mathbf{v}$ and try again.

Based on this description, the UOV scheme can be generally described by four parameters n, m, o, q . Therefore, we denote by $\text{UOV}(n, m, o, q)$ the UOV scheme with n variables and m equations over $\mathbb{F}_q$, where the oil space $\mathcal{O}$ satisfies $\dim(\mathcal{O}) = o$.

3.2 The MAYO Signature Scheme

3.2.1 The Whipping Technique

The whipping technique was first proposed by Beullens in [15] to construct the MAYO scheme. This technique allows UOV with $o < m$ to still efficiently generate signatures. Let $\mathcal{P} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ be the public map of $\text{UOV}(n, m, o, q)$ with $o < m$. Let k be a positive integer such that $ko \geq m$. The whipping technique constructs a map $\mathcal{P}^* : \mathbb{F}_q^{kn} \rightarrow \mathbb{F}_q^m$ that can be signed as follows:

$$\mathcal{P}^*(\mathbf{x}_1, \dots, \mathbf{x}_k) = \sum_{i=1}^k \mathbf{E}_{i,i} \mathcal{P}(\mathbf{x}_i) + \sum_{i=1}^k \sum_{j=i+1}^k \mathbf{E}_{i,j} \mathcal{P}'(\mathbf{x}_i, \mathbf{x}_j). \quad (1)$$

Here, $\mathbf{E}_{i,j}$ are called the emulsifier matrices of size $m \times m$, corresponding to the elements $1, z, z^2, \dots, z^{\binom{k+1}{2}-1}$ in $\mathbb{F}_q[z]/f(z)$ where $f(z)$ is an irreducible polynomial of degree m . Since there are only m basis elements in $\mathbb{F}_q[z]/f(z)$, we require $m \geq k(k+1)/2$. The reason for choosing these special matrices is that any linear combination of them is full rank.

To generate a signature for a message M , we randomly choose $(\mathbf{v}_1, \dots, \mathbf{v}_k) \in \mathbb{F}_q^{kn}$ and solve for $(\mathbf{o}_1, \dots, \mathbf{o}_k) \in \mathcal{O}^k$ such that

$$\mathcal{P}^*(\mathbf{v}_1 + \mathbf{o}_1, \dots, \mathbf{v}_k + \mathbf{o}_k) = \mathbf{t} = H(M \| \text{salt}).$$

Since $ko \geq m$, the resulting linear system has a solution with high probability.

To verify a signature, we simply check $\mathcal{P}^*(\mathbf{s}) = \mathbf{t}$ where $\mathbf{s} = (\mathbf{s}_1 \| \mathbf{s}_2 \| \dots \| \mathbf{s}_k)$ and $\{\mathbf{s}_i = \mathbf{v}_i + \mathbf{o}_i\}_{i \in [k]}$.

Regarding the key generation algorithm, MAYO adopts the key generation technique of Petzoldt [10]. Specifically, the matrices representing the map $\mathcal{P}$ are

$$\mathbf{P}_i = \begin{pmatrix} \mathbf{P}_i^{(1)} & \mathbf{P}_i^{(2)} \\ 0 & \mathbf{P}_i^{(3)} \end{pmatrix}, \quad (1 \leq i \leq m)$$

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KeyGen():
1:  $\mathbf{O} \leftarrow \mathbb{F}_q^{o \times (n-o)}$ 
2:  $\text{seed} \leftarrow \{0, 1\}^{\text{pk\_seed\_length}}$ 
3: for  $i = 1$  to  $m$  do
4:    $\mathbf{P}_i^{(1)} \leftarrow \text{Expand}(\text{seed} \parallel \text{P1} \parallel i)$   $\triangleright$  upper triangular  $(n-o) \times (n-o)$  matrix
5:    $\mathbf{P}_i^{(2)} \leftarrow \text{Expand}(\text{seed} \parallel \text{P2} \parallel i)$   $\triangleright o \times (n-o)$  matrix
6:    $\mathbf{P}_i^{(3)} \leftarrow \text{Upper}(-\mathbf{O}\mathbf{P}_i^{(1)}\mathbf{O}^T - \mathbf{O}\mathbf{P}_i^{(2)})$ 
7: end for
8: return  $(\text{pk}, \text{sk}) = ((\text{seed}, \{\mathbf{P}_i^{(3)}\}_{i \in [m]}), (\text{seed}, \mathbf{O}))$ 

Sign $(M, \text{sk})$ :
1:  $(\text{seed}, \mathbf{O}) \leftarrow \text{sk}$ 
2:  $\text{salt} \leftarrow \{0, 1\}^{\text{salt\_length}}$ 
3:  $\mathbf{t} \leftarrow \text{Hash}(M \parallel \text{salt}) \in \mathbb{F}_q^m$ 
4:  $\mathcal{P}^*(\mathbf{x}_1, \dots, \mathbf{x}_k) \leftarrow \sum_{i=1}^k \mathbf{E}_{i,i} \mathcal{P}(\mathbf{x}_i) + \sum_{1 \leq i < j \leq k} \mathbf{E}_{i,j} \mathcal{P}'(\mathbf{x}_i, \mathbf{x}_j)$ 
5:  $\mathbf{v}_i \leftarrow \mathbb{F}_q^{n-o} \times \{0\}^o$ 
6: if  $\mathcal{P}^*(\mathbf{v}_1 + \mathbf{o}_1, \dots, \mathbf{v}_k + \mathbf{o}_k)$  does not have full rank then
7:   go to Step 5
8: end if
9: Solve  $\mathcal{P}^*(\mathbf{v}_1 + \mathbf{o}_1, \dots, \mathbf{v}_k + \mathbf{o}_k) = \mathbf{t}$  for  $\mathbf{o}_1, \dots, \mathbf{o}_k \in \text{RowSpace}((\mathbf{O} \mathbf{I}_o))$ 
10: return  $\sigma = (\text{salt}, \{\mathbf{s}_i = \mathbf{v}_i + \mathbf{o}_i\}_{i \in [k]})$ 

Verify $(M, \text{pk}, \sigma)$ :
1:  $(\text{salt}, \{\mathbf{s}_i\}_{i \in [k]}) \leftarrow \sigma$ 
2:  $\mathbf{t} \leftarrow \text{Hash}(M \parallel \text{salt}) \in \mathbb{F}_q^m$ 
3:  $\mathbf{t}' \leftarrow \sum_{i=1}^k \mathbf{E}_{i,i} \mathcal{P}(\mathbf{s}_i) + \sum_{1 \leq i < j \leq k} \mathbf{E}_{i,j} \mathcal{P}'(\mathbf{s}_i, \mathbf{s}_j)$ 
4: return accept if  $\mathbf{t} = \mathbf{t}'$  and reject otherwise

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Figure 1: Key generation, signing and verification of MAYO

where $\mathbf{P}_i^{(1)}$ and $\mathbf{P}_i^{(3)}$ are upper triangular matrices of sizes $n-o$ and o , respectively, and $\mathbf{P}_i^{(2)}$ are rectangular matrices of size $(n-o) \times o$. The matrices $\mathbf{P}_i^{(1)}$ and $\mathbf{P}_i^{(2)}$ are generated from a public seed, and $\mathbf{P}_i^{(3)}$ is then computed so that the resulting map $\mathcal{P}$ vanishes on the oil space $\mathcal{O}$. Details of the key generation, signing, and verification algorithms can be found in Figure 1.

3.2.2 Key and Signature sizes

Since the public key of MAYO is exactly the map $\mathcal{P}$, by using the key generation technique of Petzoldt for UOV(n, m, o, q) as described in Figure 1, the public key size is

$$m \binom{o+1}{2} \frac{\log_2 q}{8} \quad (\text{bytes}).$$

The size of the signature $\mathbf{s}$ depends on the whipping parameter k and is given by

$$\frac{nk \log_2(q)}{8} \quad (\text{bytes}).$$

4 Security Analysis of MAYO

In this section, we present a security evaluation of MAYO. To bring all costs to a common metric, we estimate the number of gates required to perform a multiplication in the finite field $\mathbb{F}_q$ as

$$\# \text{gates} = \# \text{field multiplications} \cdot (2(\log_2 q)^2 + \log_2 q)$$

4.1 Forgery Attacks

Forgery attacks are attacks that attempt to find a preimage of the MAYO map $\mathcal{P}^*$, that is, to find $\mathbf{s}$ such that

$$\mathcal{P}^*(\mathbf{s}) = \mathbf{t} = H(M || \text{salt})$$

4.1.1 Direct Attack 1

Direct attack 1 is an attack that solves the system $\mathcal{P}^*(\mathbf{s}) = \mathbf{t}$ without considering the structure of the MAYO cryptosystem. The difficulty of this attack is equivalent to solving the MQ problem with kn variables and m equations over $\mathbb{F}_q$. The cost of this attack in this case is

$$MQ(kn, m, q)$$

4.1.2 Direct Attack 2

The attacker can also exploit the rank property of the matrices $\mathbf{E}_{i,j}$ by performing a change of variables $\mathbf{x}_i = \alpha_i \mathbf{x}_1 + \beta_i \forall 2 \leq i \leq k$ with $\alpha_i, \beta_i \in \mathbb{F}_q$ to obtain the equation

$$\mathbf{E}_\alpha \mathcal{P}(\mathbf{x}_1) + \text{Affine function of } \mathbf{x}_1 = \mathbf{t}^* \quad (2)$$

where $\mathbf{E}_\alpha = \sum_{i=1}^k \alpha_i^2 \mathbf{E}_{i,i} + \sum_{i=1}^k \sum_{j=i+1}^k \alpha_i \alpha_j \mathbf{E}_{i,j}$ with $\alpha_1 = 1$.

Since the matrices $\mathbf{E}_{i,j}$ are designed such that every linear combination of them has full rank, $\mathbf{E}_\alpha$ has full rank. Therefore, in this case we obtain a system of m quadratic multivariate equations in n variables. The cost of this attack is

$$MQ(n, m, q)$$

Since solving the MQ system in this case has the same number of equations as Direct Attack 1 but does not benefit from algorithms such as Hashimoto's, its cost is always greater than that of Direct Attack 1.

4.1.3 Collision Attack

This is an attack where the attacker tries to find a collision of the equation $\mathcal{P}^*(\mathbf{s}) = H(M || \text{salt})$. The attacker can compute $\mathcal{P}^*(\mathbf{s}^{(i)})$ for X inputs $\{\mathbf{s}^{(i)}\}_{i \in [X]}$ and compute $H(M || \text{salt}_j)$ for Y values $\{\text{salt}_j\}_{j \in [Y]}$. If $XY = \alpha q^m$, then a collision will occur with probability $\approx 1 - e^{-\alpha}$ and the attacker can forge a signature for the message M . On the other hand, $\mathcal{P}^*$ is a homogeneous system, so $\mathcal{P}^*(\sqrt{c}\mathbf{s}) = c\mathcal{P}^*(\mathbf{s})$, therefore assuming we are working in a field of even characteristic, we have that X evaluations of $\mathcal{P}^*(\mathbf{s}^{(i)})$ yield $(q-1)X$ values. Consequently, we can achieve the same probability $1 - e^{-\alpha}$ with $XY = \frac{\alpha q^m}{q-1}$.

Let ϵ be the cost of one evaluation of $\mathcal{P}^*(\mathbf{s})$ and λ be the cost of one hash computation. Then the cost of the attack is

$$\frac{1}{1 - e^{-\alpha}} (\epsilon X + \lambda Y) = \frac{1}{1 - e^{-\alpha}} \left(\epsilon X + \lambda \frac{\alpha q^m}{X(q-1)} \right) \geq \frac{2\sqrt{\alpha}}{1 - e^{-\alpha}} \sqrt{\epsilon \lambda \frac{q^m}{q-1}}$$

According to [29], we have $\lambda = 2^{17.5}$. On the other hand, by using Gray-code enumeration [30] and evaluating only the first $m/2$ equations to find a collision, we obtain $\epsilon = \frac{3m \log_2(q)}{2}$. By choosing $\alpha = 1.25$ to minimize $\frac{2\sqrt{\alpha}}{1-e^{-\alpha}}$, we get the cost of the attack as

$$ColAtk(m, q) := 2^{10.7} \sqrt{m \frac{q^m}{q-1} \log_2(q)}$$

For quantum collision attacks, it has been shown that an attacker with finite time would prefer classical attacks as in [31]. Therefore, in the following, the estimates for both cases will use classical estimates.

4.2 Key-Recovery Attacks

Key-recovery attack in the MAYO scheme is equivalent to performing key-recovery attack for the $UOV(n, m, o, q)$ scheme by exploiting the structure of the public key $\mathcal{P}$. Therefore, in this section, we will examine key-recovery attacks on $UOV(n, m, o, q)$.

From [32], we know that the entire oil space can be recovered in polynomial time given just one non-zero vector contained in it. On the other hand, let $\mathcal{O}^\perp$ be the vinegar space. Clearly, we can also recover the oil space $\mathcal{O}$ if $\mathcal{O}^\perp$ is known. Consequently, the goal of key-recovery attacks is to find at least one non-zero vector lying in the oil space $\mathcal{O}$ or to recover the vinegar space $\mathcal{O}^\perp$.

4.2.1 Kipnis–Shamir Attack

The attack was introduced in 1998 by Kipnis and Shamir [8]. The attack attempts to find vectors in the oil space $\mathcal{O}$ by exploiting the fact that these vectors are likely to be eigenvectors of the matrices $\mathbf{W} = \mathbf{W}_i^{-1} \mathbf{W}_j$, where $\mathbf{W}_i$ and $\mathbf{W}_j$ are matrices formed by linear combinations of $\mathbf{P}_1, \dots, \mathbf{P}_m$.

According to [28], one must compute the characteristic polynomial of approximately q^{n-2o-1} matrices $\mathbf{W}$ of size $n \times n$ and solve the same number of systems of linear equations in n variables. Therefore, the cost of the attack can be estimated as

$$q^{n-2o-1} n^{2.8} (2 \log_2(q)^2 + \log_2(q))$$

Grover’s algorithm [24] can be used here to reduce the complexity to

$$q^{\frac{n-2o-1}{2}} n^{2.8} (2 \log_2(q)^2 + \log_2(q))$$

4.2.2 Reconciliation Attack

The reconciliation attack was proposed by Ding [33]. This attack exploits the property $\mathcal{P}(\mathbf{o}) = 0$ for all $\mathbf{o} \in \mathcal{O}$ to find a non-zero oil vector in $\mathcal{O}$. Specifically, we impose o affine constraints on the system $\mathcal{P}(\mathbf{o}) = 0$. This results in a quadratic multivariate system consisting of m equations and $n - o$ variables.

If $n - o \leq m$, this system with high probability has only one solution. Therefore, by solving the MQ problem with $n - o$ variables and m equations over $\mathbb{F}_q$, we can recover $\mathbf{o}$ and hence reconstruct the oil space $\mathcal{O}$. The cost in this case is

$$MQ(n - o, m, q)$$

If $n - o > m$, the system has approximately q^{n-o-m} solutions, and only one of them belongs to the oil space. In this case, there are two strategies to find this oil vector. The first strategy is

to guess variables and repeatedly solve the system with $n - o$ variables and m equations over $\mathbb{F}_q$ about q^{n-o-m} times. The second strategy is to find multiple oil vectors simultaneously by solving the system

$$\begin{cases} \mathcal{P}(\mathbf{o}_i) = 0 & \forall i \in \{1, \dots, k\} \\ \mathcal{P}'(\mathbf{o}_i, \mathbf{o}_j) = 0 & \forall i < j \in \{1, \dots, k\} \end{cases}$$

Clearly, the first strategy is more efficient when q^{n-o-m} is small, and vice versa. By combining the two strategies, the cost of the attack in this case is

$$\min \left\{ MQ \left(k_{\min}(n - o), \frac{k_{\min}(k_{\min} + 1)}{2}m, q \right), q^{c(n-o) - \frac{c(c+1)}{2}m} MQ \left(\frac{c(c+1)}{2}m, \frac{c(c+1)}{2}m, q \right) \right\}$$

where $k_{\min} = \min_{k \in \mathbb{Z}} \left\{ k \mid k \geq \frac{2(n-o)-m}{m} \right\}$ and $c = k_{\min} - 1$.

By applying Grover's algorithm, the quantum cost can be estimated as

$$\min \left\{ MQ \left(k_{\min}(n - o), \frac{k_{\min}(k_{\min} + 1)}{2}m, q \right), \sqrt{q^{c(n-o) - \frac{c(c+1)}{2}m}} MQ \left(\frac{c(c+1)}{2}m, \frac{c(c+1)}{2}m, q \right) \right\}$$

4.2.3 Intersection Attack

In [28], Beullens proposed the Intersection Attack, which combines the Kipnis–Shamir attack and the Reconciliation attack.

In the case $n < 3o$, let $k \geq 2$ be an integer satisfying $k < \frac{n-o}{n-2o}$. Let $\mathbf{W}_1, \dots, \mathbf{W}_k$ be k invertible matrices randomly chosen from the linear combinations of the matrices representing the public key $\mathbf{P}_1, \dots, \mathbf{P}_m$. The attack attempts to solve the following system of equations:

$$\begin{cases} \mathcal{P}(\mathbf{W}_i^{-1}\mathbf{x}) = 0 & \forall i \in \{1, \dots, k\} \\ \mathcal{P}'(\mathbf{W}_i^{-1}\mathbf{x}, \mathbf{W}_j^{-1}\mathbf{y}) = 0 & \forall i < j \in \{1, \dots, k\} \end{cases}$$

According to [28], this system consists of $\binom{k+1}{2}m - 2\binom{k}{2}$ equations and $kn - (2k - 1)o$ variables. Therefore, the cost in this case is

$$MQ \left(kn - (2k - 1)o, \binom{k+1}{2}m - \binom{k}{2}, q \right)$$

In the case $n \geq 3o$, with $k = 2$ the attack succeeds with probability q^{n-3o+1} . Therefore, the cost in this case is

$$q^{n-3o+1} MQ(n, 3m - 2, q)$$

The quantum estimate obtained by applying Grover's algorithm is

$$q^{\frac{n-3o+1}{2}} MQ(n, 3m - 2, q)$$

4.2.4 Rectangular MinRank Attack

The rectangular MinRank Attack was first proposed by Beullens in [28] to attack Rainbow. Later, Furue and Ikematsu [34] showed that this attack can also be applied to $\text{UOV}(n, m, o, q)$ when $n - o < m$.

Let

$$(\tilde{\mathbf{P}}_1, \dots, \tilde{\mathbf{P}}_n) := \text{Rect}_{\text{col}}(\mathbf{P}_1, \dots, \mathbf{P}_m)$$

$$(\tilde{\mathbf{F}}_1, \dots, \tilde{\mathbf{F}}_n) := \text{Rect}_{\text{col}}(\mathbf{F}_1, \dots, \mathbf{F}_m)$$

From the definition of the rectangular transformation, we always have

$$(\tilde{\mathbf{P}}_1, \dots, \tilde{\mathbf{P}}_n) = \left(\mathbf{T}^\top \tilde{\mathbf{F}}_1, \dots, \mathbf{T}^\top \tilde{\mathbf{F}}_n \right) \mathbf{T}.$$

Therefore, for $\mathbf{o} \in \mathcal{O}$ we always have the following system:

$$\begin{cases} \mathcal{P}(\mathbf{o}) = 0, \\ \text{rank}(\text{Rect}_{\text{col}}(\mathbf{P}_1, \dots, \mathbf{P}_m) \cdot \mathbf{o}) \leq n - o. \end{cases}$$

If $n - o \geq m$, then $\text{rank}(\text{Rect}_{\text{col}}(\mathbf{P}_1, \dots, \mathbf{P}_m) \cdot \mathbf{o}) \leq n - o \quad \forall \mathbf{o} \in \mathbb{F}_q^n$. Therefore, the attack is only meaningful when $n - o < m$.

On the other hand, by applying the support minor modeling method [35], the cost of this attack is

$$\min_{m' \in [n-o+1, m]} \left\{ 3 \binom{m'}{n-o}^2 \binom{n-o+b_{\min}}{b_{\min}}^2 (n-o+1)^2 \right\} (2 \log_2(q)^2 + \log_2(q))$$

where the value $b_{\min}$ is determined as follows.

Define $R'(b)$ and $G(t_1, t_2)$ by

$$R'(b) := \sum_{i=1}^b (-1)^{i+1} \binom{m'}{n-o+i} \binom{n+i-1}{i} \binom{n-o+b-i}{b-i}$$

$$G(t_1, t_2) := (1 - t_1^2)^m \left(\binom{m'}{n-o} t_2 + \sum_{b=1}^{n-o+1} \left(\binom{m'}{n-o} \binom{n-o+b-1}{b} - R'(b) \right) t_1^b t_2 \right)$$

Let $G(t_1, t_2)_{b,1}$ denote the coefficient of $t_1^b t_2$. Then we define

$$b_{\min} := \min\{b \in \mathbb{N} \mid G(t_1, t_2)_{b,1} \leq 1\}.$$

In this case, Grover's quantum algorithm does not provide any advantage, so the classical and quantum estimates are the same.

4.2.5 Wedge Attack

The Wedge Attack is an attack that exploits properties of exterior algebra to recover the vinegar space $\mathcal{O}^\perp$. It was proposed by Lan Ran in [36] for fields of even characteristic, and a generalized version was later presented in [37]. Since we use a field of even characteristic in the following sections, we only focus on Lan Ran's Wedge Attack here.

Consider the linear map $\mathcal{M}$:

$$\mathcal{M} : \bigwedge^{n-o} \mathbb{F}_q^n \longrightarrow \left(\bigwedge^{n-o+2} \mathbb{F}_q^n \right)^m$$

$$\alpha \longmapsto (\alpha \wedge p'_1, \dots, \alpha \wedge p'_m),$$

where $\wedge$ denotes the wedge product. More details can be found in [38].

In [36], Lan Ran showed that the vinegar space $\mathcal{O}^\perp$ with basis $(v_1, \dots, v_{n-o})$ has an associated element $v_1 \wedge v_2 \wedge \dots \wedge v_{n-o}$ that lies in the kernel of $\mathcal{M}$. When $\ker(\mathcal{M})$ has dimension 1, we can easily recover $v_1 \wedge v_2 \wedge \dots \wedge v_{n-o}$ and therefore the vinegar space $\mathcal{O}^\perp$.

Therefore, the cost of this attack mainly depends on finding the kernel of $\mathcal{M}$. On the other hand, to optimize the size of the matrix representing $\mathcal{M}$, we apply a projection that reduces the problem $\text{UOV}(n, m, o, q)$ to $\text{UOV}(n', m, o', q)$ with $o' < o$. By using the Block Wiedemann algorithm [39], the cost of the attack is

$$3 \binom{n-o+2}{2} \binom{n-o+o'}{n-o}^2 (2 \log_2(q)^2 + \log_2(q))$$

where

$$o' = \min \left\{ o' \left| \frac{2(n-o)}{m} + 1 < o' < o, \sum_{i=0}^{\lfloor \frac{o'}{2} \rfloor} (-1)^i \binom{m+i-1}{i} \binom{n-o+o'}{n-o+2i} \leq 1 \right. \right\}$$

is the condition ensuring that the matrix $\mathcal{M}$ has a one-dimensional kernel.

As in the Rectangular MinRank Attack, the classical and quantum estimates are the same in this case.

5 Solving the Lifted Multivariate Quadratic System

In this section, we present existing algorithms for solving the lifting system corresponding to the map $\mathcal{P} : \mathbb{F}_{q^r}^n \rightarrow \mathbb{F}_{q^r}^m$.

$$\mathcal{P}(\mathbf{x}) = \begin{cases} p_1(\mathbf{x}) = \sum_{i=1}^n \sum_{j=i}^n a_{i,j,1} x_i x_j + \sum_{i=1}^n b_{i,1} x_i = y_1 \\ p_2(\mathbf{x}) = \sum_{i=1}^n \sum_{j=i}^n a_{i,j,2} x_i x_j + \sum_{i=1}^n b_{i,2} x_i = y_2 \\ \vdots \\ p_m(\mathbf{x}) = \sum_{i=1}^n \sum_{j=i}^n a_{i,j,m} x_i x_j + \sum_{i=1}^n b_{i,m} x_i = y_m \end{cases}$$

where $a_{i,j,k}$ and $b_{i,k}$ are sampled uniformly at random from $\mathbb{F}_q$.

5.1 Subfield Differential Attack

SDA was first introduced by Ding in [27] to attack LUOV round 1. Let d be a positive integer dividing r . We perform the change of variables $\mathbf{x} = \bar{\mathbf{x}} + \mathbf{x}'$ with $\bar{\mathbf{x}} \in \mathbb{F}_{q^d}^n$ and $\mathbf{x}'$ a random value sampled from $\mathbb{F}_{q^r}^n$. By applying this variable transformation, we obtain

$$\tilde{\mathcal{P}} : \mathbb{F}_{q^d}^n \rightarrow \mathbb{F}_{q^r}^m, \quad \tilde{\mathcal{P}}(\bar{\mathbf{x}}) = (\tilde{p}_1(\bar{\mathbf{x}}), \tilde{p}_2(\bar{\mathbf{x}}), \dots, \tilde{p}_m(\bar{\mathbf{x}})) := \mathcal{P}(\bar{\mathbf{x}} + \mathbf{x}').$$

On the other hand, we have

$$\mathbb{F}_{q^r} \cong \mathbb{F}_{q^d}[t]/g(t)$$

where $g(t)$ is an irreducible polynomial of degree $s = r/d$.

Therefore, by representing $\mathbf{y} = (y_1, \dots, y_m)$ and $\tilde{\mathcal{P}}$ over $\mathbb{F}_{q^d}$, we obtain

$$\begin{aligned}\tilde{p}_1(\bar{\mathbf{x}}) &= Q_1(\bar{\mathbf{x}}) + \sum_{i=1}^{s-1} L_{i,1}(\bar{\mathbf{x}})t^i = y_{0,1} + \sum_{i=1}^{s-1} y_{i,1}t^i \\ \tilde{p}_2(\bar{\mathbf{x}}) &= Q_2(\bar{\mathbf{x}}) + \sum_{i=1}^{s-1} L_{i,2}(\bar{\mathbf{x}})t^i = y_{0,2} + \sum_{i=1}^{s-1} y_{i,2}t^i \\ &\vdots \\ \tilde{p}_m(\bar{\mathbf{x}}) &= Q_m(\bar{\mathbf{x}}) + \sum_{i=1}^{s-1} L_{i,m}(\bar{\mathbf{x}})t^i = y_{0,m} + \sum_{i=1}^{s-1} y_{i,m}t^i\end{aligned}$$

where each Q_k is a quadratic polynomial over $\mathbb{F}_{q^d}$ and each $L_{i,k}$ is a linear polynomial over $\mathbb{F}_{q^d}$.

To solve this system, we first compute the solution space A of the system consisting of $(s-1)m$ linear equations in n variables

$$\mathcal{L} = \{ L_{i,j}(\bar{\mathbf{x}}) = y_{i,j} \mid 1 \leq i \leq s-1, 1 \leq j \leq m \}.$$

Since $\mathcal{L}$ is a random linear system, with high probability the dimension of A is

$$\dim(A) = \max\{n - (s-1)m, 0\}.$$

Next, we solve the quadratic system consisting of m equations under the constraint that the solution lies in A

$$\mathcal{Q} = \{ Q_i(\bar{\mathbf{x}}) = y_{0,i} \mid 1 \leq i \leq m, \bar{\mathbf{x}} \in A \}.$$

According to Lemma 1, this system has a solution with probability $1 - e^{-q^{d(n-(s-1)m-m)}}$. Since this probability decreases exponentially fast, we expect the algorithm to succeed if and only if $n - (s-1)m > m$, that is, at least $n/m > s$ is required.

5.2 Nested Subset Differential Attack

NSDA is an improvement of SDA proposed by Ding in [12] to attack LUOV round 2. This algorithm results in a final system with fewer linear equations and therefore increases the applicability range of the attack. Let $\mu = \lceil (s-1)/2 \rceil$ and $\mathcal{H} = \{\mu, \mu+1, \dots, 2\mu\}$. The attack finds a solution $\mathbf{a} = a_0 + a_1t + \dots + a_\mu t^\mu$ by solving the following systems sequentially

$$\begin{aligned}\bar{\mathcal{P}}^0(a_0) &= y_0 \\ \bar{\mathcal{P}}^1(a_0 + a_1t) &= y_1t \\ &\vdots \\ \bar{\mathcal{P}}^{\mu-1}(a_0 + \dots + a_{\mu-1}t^{\mu-1}) &= y_{\mu-1}t^{\mu-1} \\ \bar{\mathcal{P}}^{\mathcal{H}}(a_0 + \dots + a_\mu t^\mu) &= \sum_{i=\mu}^{2\mu-1} y_i t^i\end{aligned}$$

According to [12], the main bottleneck in solving this system lies in the last expression, which has the following form

$$\begin{aligned}
\bar{p}_1^{\mathcal{H}}(a_0 + \dots + a_{\mu-1}t^{\mu-1} + \bar{\mathbf{x}}t^{\mu}) &= \sum_{i=\mu}^{2\mu-1} L_{i,1}(\bar{\mathbf{x}})t^i + Q_{2\mu,1}(\bar{\mathbf{x}})t^{2\mu} = \sum_{i=\mu}^{2\mu} y_{i,1}t^i \\
\bar{p}_2^{\mathcal{H}}(a_0 + \dots + a_{\mu-1}t^{\mu-1} + \bar{\mathbf{x}}t^{\mu}) &= \sum_{i=\mu}^{2\mu-1} L_{i,2}(\bar{\mathbf{x}})t^i + Q_{2\mu,2}(\bar{\mathbf{x}})t^{2\mu} = \sum_{i=\mu}^{2\mu} y_{i,2}t^i \\
&\vdots \\
\bar{p}_m^{\mathcal{H}}(a_0 + \dots + a_{\mu-1}t^{\mu-1} + \bar{\mathbf{x}}t^{\mu}) &= \sum_{i=\mu}^{2\mu-1} L_{i,m}(\bar{\mathbf{x}})t^i + Q_{2\mu,m}(\bar{\mathbf{x}})t^{2\mu} = \sum_{i=\mu}^{2\mu} y_{i,m}t^i
\end{aligned}$$

Similarly to SDA, the solution space A has dimension

$$\dim(A) = \max\{n - \mu m, 0\}.$$

Therefore, the probability that this system has a solution is $1 - e^{q^{-d(n-\mu m-m)}}$, which means that at least $n/m > \mu + 1 = \lceil (s-1)/2 \rceil + 1$ is required.

5.3 NSDA+

NSDA+ is an improved version of NSDA presented in [40]. This algorithm combines the ideas of SDA and NSDA. The attack first performs a variable transformation as in SDA to obtain a system with a better form, and then solves a sequence of systems of equations as in NSDA.

For this algorithm, we have

$$\dim(A) = \max\{n - (\mu + \nu)m, 0\}$$

where $\mu, \nu \in \mathbb{N}$ satisfy $2\mu + \mu\nu + \nu \geq s - 1$.

Therefore, the probability that the algorithm succeeds is $1 - e^{q^{-d(n-(\mu+\nu)m-m)}}$, which means that at least $n/m > \mu + \nu + 1$ is required. More details about the attack can be found in [40].

6 Improvements to the Whipping Technique

6.1 Lifting Technique for Emulsifier Matrices

The lifting technique, first introduced by Beullens in [11] to construct the LUOV scheme, uses a lifting parameter r . This technique generates keys over the base field $\mathbb{F}_q$, while signing and verification are performed over the extension field $\mathbb{F}_{q^r}$. As a result, the technique reduces the key size at the cost of increasing the signature size.

Although the lifting technique has been shown to be insecure due to the lifted system-solving attacks discussed in the previous section, we observe that these attacks require a sufficiently large ratio between the number of variables and the number of equations in order to succeed. On the other hand, by using the whipping technique, this ratio can be kept close to one and therefore cannot be exploited by such attacks.

More specifically, we combine lifting with whipping by selecting emulsifier matrices $\mathbf{E}_{i,j}^{(L)}$ whose entries lie in $\mathbb{F}_{q^r}$ instead of $\mathbb{F}_q$, while keeping the underlying public UOV maps defined over $\mathbb{F}_q$. These matrices are still chosen so as to preserve the property that any linear combination of them has full rank, meaning that the matrices $\mathbf{E}_{i,j}^{(L)}$ correspond to basis elements of $\mathbb{F}_{(q^r)^m}$. Selecting

emulsifier matrices over the extension field produces a map $\mathcal{P}^* : \mathbb{F}_{q^r}^{kn} \rightarrow \mathbb{F}_{q^r}^m$ which is a system of multivariate quadratic polynomials with coefficients in $\mathbb{F}_{q^r}$. Consequently, lifted system-solving attacks cannot be applied. The only remaining way for an attacker to exploit lifted-system solving attacks is to target a smaller subsystem, as in Direct Attack 2.

From Equation (1), each component of $\mathcal{P}^*$ can be written as

$$p_l^*(\mathbf{x}_1, \dots, \mathbf{x}_k) = \sum_{i=1}^k \sum_{h=1}^m \varepsilon_{ih}^{(l)} p_h(\mathbf{x}_i) + \sum_{1 \leq i \leq j \leq k} \sum_{h=1}^m \eta_{ijh}^{(l)} p_h'(\mathbf{x}_i, \mathbf{x}_j)$$

where $\varepsilon_{ih}^{(l)}, \eta_{ijh}^{(l)} \in \mathbb{F}_{q^r}$ are coefficients taken from the l -th row of $\mathbf{E}_{i,i}^{(L)}$ and $\mathbf{E}_{i,j}^{(L)}$, respectively.

Let the matrix representations of p_i and p_i' be denoted by $\mathbf{P}_i$ and $\mathbf{P}_i'$. Then p_l^* can be written as

$$p_l^*(\mathbf{x}_1, \dots, \mathbf{x}_k) = \mathbf{x}^T \mathbf{P}_l^* \mathbf{x}$$

where

$$\mathbf{P}_l^* = \begin{pmatrix} \sum_{h=1}^m \varepsilon_{1h}^{(l)} \mathbf{P}_h & \sum_{h=1}^m \eta_{12h}^{(l)} \mathbf{P}_h' & \cdots & \sum_{h=1}^m \eta_{1kh}^{(l)} \mathbf{P}_h' \\ 0 & \sum_{h=1}^m \varepsilon_{2h}^{(l)} \mathbf{P}_h & \cdots & \sum_{h=1}^m \eta_{2kh}^{(l)} \mathbf{P}_h' \\ 0 & 0 & \ddots & \vdots \\ 0 & 0 & 0 & \sum_{h=1}^m \varepsilon_{kh}^{(l)} \mathbf{P}_h \end{pmatrix}.$$

Since the matrices $\mathbf{P}_h$ and $\mathbf{P}_h'$ are defined over $\mathbb{F}_q$ while the coefficients ε and η lie in $\mathbb{F}_{q^r}$, the matrix $\mathbf{P}_l^*$ has entries in $\mathbb{F}_{q^r}$. Therefore, $\mathcal{P}^*$ is a system of m quadratic equations over $\mathbb{F}_{q^r}$ in kn variables.

Next, the signing algorithm of MAYO in this setting is based on finding vectors $\mathbf{o}_i \in (\mathcal{O} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^r}) \subset \mathbb{F}_{q^r}^n$ such that

$$\mathcal{P}^*(\mathbf{v}_1 + \mathbf{o}_1, \dots, \mathbf{v}_k + \mathbf{o}_k) = \mathbf{t},$$

where $\mathbf{v}_i \in \mathbb{F}_{q^r}^n$ are chosen uniformly at random and

$$\mathbf{t} = \text{Hash}(M || \text{salt}) \in \mathbb{F}_{q^r}^m.$$

Using the properties $\mathcal{P}(\mathbf{o}) = 0$ and $\mathcal{P}'(\mathbf{o}_i, \mathbf{o}_j) = 0$, we obtain

$$\mathcal{P}^*(\mathbf{v} + \mathbf{o}) = \underbrace{\sum_{i=1}^k \mathbf{E}_{ii}^{(L)} \mathcal{P}(\mathbf{v}_i) + \sum_{i < j} \mathbf{E}_{ij}^{(L)} \mathcal{P}'(\mathbf{v}_i, \mathbf{v}_j)}_{\text{fixed by the choice of } \mathbf{v}} + \underbrace{\sum_{i=1}^k \mathbf{E}_{ii}^{(L)} \mathcal{P}'(\mathbf{v}_i, \mathbf{o}_i) + \sum_{i < j} \mathbf{E}_{ij}^{(L)} (\mathcal{P}'(\mathbf{v}_i, \mathbf{o}_j) + \mathcal{P}'(\mathbf{o}_i, \mathbf{v}_j))}_{\text{linear function of } \mathbf{o}}.$$

Thus, we obtain a linear system with ko variables and m equations over $\mathbb{F}_{q^r}$. Since $ko \geq m$, this system has a solution with high probability. Empirically, we observe that the probability of signing failure is approximately $q^{r(m-ko-1)}$.

6.1.1 Choice of Subfield and Lifting Parameter

In this section, we consider the choice of subfield as well as the lifting parameter when combined with the whipping technique, in order to obtain the most secure and efficient scheme.

We first discuss the lifting parameter r . Since the whipping technique increases the signature size by a factor of k , it is not practical to choose a large lifting parameter r . On the other hand, we aim to ensure that the success probabilities of SDA, NSDA, and NSDA+ remain essentially

the same; in other words, improvements along this line of attacks should not lead to a significant advantage. For this reason, we consider $r = 2$ to be the most suitable choice.

Next, we turn to the choice of the subfield. We prioritize subfields of even characteristic, as they are more implementation-friendly. In addition, we avoid choosing very small base fields, since there exist algorithms that solve MQ systems faster than WXL over small fields. For instance, the algorithm of Dinur [41] is particularly efficient for systems over $\mathbb{F}_2$, while the Crossbred algorithm [42] performs well over small fields. To illustrate this, we compare their complexities in Table 1. The cost estimates are obtained using the CryptographicEstimators library [43].

From Table 1, we observe that over the small field $q = 2$, the Dinur-II algorithm is consistently faster than WXL. Over $q = 4$, the Crossbred algorithm becomes dominant. However, for larger fields with $q \geq 8$, WXL becomes more efficient overall. Although Crossbred may exhibit slightly lower asymptotic time complexity when $q = 8$, it incurs significantly higher memory consumption than WXL. In practice, this leads to substantial overhead, as memory access costs dominate arithmetic operations when memory usage is large. Consequently, the real-world running time of Crossbred in this setting is comparable to, or even exceeds, that of WXL.

Based on the above analysis, we propose using fields with $q = 8$ and $q = 16$ when combining the lifting and whipping techniques.

Table 1: Time and memory complexities of WXL, Dinur-II, and Crossbred algorithms over $\mathbb{F}_{2^k}$ for $n = m \in \{60, 80, 100\}$. The best values in each case are highlighted in bold.

		$n = m$					
		60		80		100	
		Time	Mem	Time	Mem	Time	Mem
$\mathbb{F}_2$	WXL	2^{65}	2^{17}	2^{85}	2^{18}	2^{105}	2^{19}
	Dinur-II	2^{59}	2^{46}	2^{75}	2^{59}	2^{92}	2^{72}
	Crossbred	2^{62}	2^{41}	2^{80}	2^{61}	2^{98}	2^{73}
$\mathbb{F}_{2^2}$	WXL	2^{114}	2^{22}	2^{146}	2^{31}	2^{176}	2^{36}
	Dinur-II	—	—	—	—	—	—
	Crossbred	2^{105}	2^{82}	2^{137}	2^{113}	2^{169}	2^{144}
$\mathbb{F}_{2^3}$	WXL	2^{140}	2^{38}	2^{179}	2^{41}	2^{218}	2^{49}
	Dinur-II	—	—	—	—	—	—
	Crossbred	2^{136}	2^{116}	2^{177}	2^{152}	2^{219}	2^{184}
$\mathbb{F}_{2^4}$	WXL	2^{157}	2^{49}	2^{200}	2^{66}	2^{243}	2^{76}
	Dinur-II	—	—	—	—	—	—
	Crossbred	2^{159}	2^{132}	2^{206}	2^{180}	2^{260}	2^{197}

6.1.2 Impact of Lifting on Existing Emulsifier Matrix Modifications

In this section, we discuss the benefits of the lifting modification when applied to previously proposed emulsifier matrix modifications.

We first consider the rectangular modification presented in [17, 18]. Instead of using square matrices of size $m \times m$ as in the original MAYO scheme, this modification employs rectangular matrices $\mathbf{E}_{i,j}^{(R)} \in \mathbb{F}_q^{m \times (m-a)}$ with $0 \leq a < m$. These matrices are structured in block form:

$$\mathbf{E}_{i,j}^{(R)} = \begin{pmatrix} \mathbf{A}_{i,j}^{(R)} \\ \mathbf{B}_{i,j}^{(R)} \end{pmatrix},$$

where $\mathbf{A}_{i,j}^{(R)}$ is a square matrix of size $(m-a) \times (m-a)$ chosen to correspond to a basis element of the extension field $\mathbb{F}_{q^{m-a}}$, and $\mathbf{B}_{i,j}^{(R)}$ are random matrices of size $a \times (m-a)$. With this construction, any linear combination of the matrices $\mathbf{E}_{i,j}^{(R)}$ remains full rank $m-a$, since the first $m-a$ rows are chosen in the same way as in MAYO.

The benefit of using rectangular emulsifier matrices is that we can now employ mini UOV instances $\text{UOV}(n, m-a, o, q)$ instead of $\text{UOV}(n, m, o, q)$ without changing the complexity of Direct Attack 1. On the other hand, this also increases the hardness of the underlying UOV problem. Consequently, this modification allows us to obtain

- Smaller public keys since the number of equations decreases.
- Smaller signatures because the key-recovery attacks become harder, allowing us to reduce the number of variables without changing the security level.
- Faster signing and verification since the matrices $\mathbf{E}_{i,j}^{(R)}$ are smaller and $\mathcal{P}$ has fewer variables and equations.

It is easy to see that choosing a larger parameter a leads to smaller public key and signature sizes. However, increasing a strengthens collision attacks and direct attack 2, as will be discussed later, making them more effective than other attacks. On the other hand, lifting the emulsifier matrices to an extension field makes signature forgery attacks more difficult. Therefore, this lifting technique allows us to safely choose a larger value of a .

To illustrate this, we present two security level I parameter sets with the same security level:

$$(n, m, o, k, q, r, a) = (62, 56, 8, 7, 16, 2, 8) \quad \text{and} \quad (n, m, o, k, q, r, a) = (80, 78, 8, 10, 16, 1, 6).$$

The corresponding public key and signature sizes are (876; 458) bytes and (1312; 424) bytes, respectively.

Next, we consider the emulsifier matrix modification proposed in [19]. The main idea of this approach is to set some matrices $\mathbf{E}_{i,j} = 0$ for $i \neq j$, in order to accelerate the signing and verification procedures without affecting the security of the MAYO scheme. However, this modification naturally leads to K -tree attacks [44] with complexity $O(K \cdot q^{m/(1+\lceil \log_2(K) \rceil)})$.

In [19], the author proposed two constructions of $\mathcal{P}^*$ depending on the whipping parameter k to mitigate the strength of K -tree attacks. Specifically, for $k \geq 8$, we have

$$\mathcal{P}^*(\mathbf{x}_1, \dots, \mathbf{x}_k) = \sum_{i=1}^k \mathbf{E}_{ii} \mathcal{P}(\mathbf{x}_i) + \sum_{i=1}^{\lfloor k/2 \rfloor} \mathbf{E}_{2i-1, 2i} \mathcal{P}'(\mathbf{x}_{2i-1}, \mathbf{x}_{2i}).$$

For $k < 8$, only the diagonal matrices $\mathbf{E}_{i,i} \neq 0$ are retained.

For example, when $k = 8$, by setting $\mathbf{x}_2 = \mathbf{x}_4 = \mathbf{x}_6 = \mathbf{x}_8 = 0$, the problem reduces to a 4-sum instance of the form

$$\mathbf{t} = \mathbf{E}_{11} \mathcal{P}(\mathbf{x}_1) + \mathbf{E}_{33} \mathcal{P}(\mathbf{x}_3) + \mathbf{E}_{55} \mathcal{P}(\mathbf{x}_5) + \mathbf{E}_{77} \mathcal{P}(\mathbf{x}_7).$$

For the security level I parameter set $(n, m, o, k, q, r) = (76, 64, 8, 8, 8, 2)$ of MAYOL (see Appendix B), the cost of the K -tree attack is 2^{130} . Similarly, for the level-1 parameter set $(n, m, o, k, q) = (86, 78, 8, 10, 16)$ of MAYO, the cost is $2^{106.32}$. Therefore, it is clear that the lifting technique improves the modification proposed in [19].

Note that although the attack costs in both cases are below the required security level, comparing them is sufficient to demonstrate the advantage of our improvement.

6.2 Enhancing Rectangular Emulsifier Matrices with Structured Designs

In this section, we first analyze the probability that the system $\mathcal{P}^*$ contains m linearly independent equations when the emulsifier matrices $\mathbf{E}_{i,j}$ are non-square. We then propose a structured choice of the matrices $\mathbf{B}_{i,j}^{(R)}$ to accelerate the signing and verification procedures, while ensuring that $\mathcal{P}^*$ still has a high probability of producing m linearly independent equations.

It is easy to see that when the emulsifier matrices are square and invertible, if the UOV map $\mathcal{P}$ is a multivariate system consisting of m linearly independent equations, then the map $\mathcal{P}^*$ will also form a multivariate system consisting of m linearly independent equations.

However, when emulsifier matrices are non-square matrices as $\mathbf{E}_{i,j}^{(R)}$, even if $\mathcal{P}$ is a multivariate system consisting of $m-a$ linearly independent equations, the resulting map $\mathcal{P}^*$ does not necessarily yield m linearly independent equations.

Lemma 2. *Let $L = k(k+1)/2$ be the number of emulsifier matrices, and let $N = kn(kn+1)/2$ be the dimension of the space generated by homogeneous quadratic monomials in kn variables. Let $\mathbf{M}_{\mathcal{P}^*} \in \mathbb{F}_q^{m \times N}$ denote the Macaulay matrix of $\mathcal{P}^*$ constructed from a UOV map $\mathcal{P}$ consisting of $m-a$ linearly independent equations and emulsifier matrices $\mathbf{E}_{i,j}^{(R)}$. Then*

$$\Pr[\text{rank}(\mathbf{M}_{\mathcal{P}^*}) = m] = \prod_{i=0}^{a-1} \left(1 - q^{(m-a+i)-L(m-a)}\right).$$

Proof. For each $i \in [k]$, let $\mathbf{M}_{i,i} \in \mathbb{F}_q^{(m-a) \times N}$ denote the Macaulay matrix of $\mathcal{P}(\mathbf{x}_i)$. For each $i < j$, let $\mathbf{M}_{i,j} \in \mathbb{F}_q^{(m-a) \times N}$ denote the Macaulay matrix of $\mathcal{P}'(\mathbf{x}_i, \mathbf{x}_j)$. These matrices have the following properties:

- $\mathbf{M}_{i,i}$ has nonzero columns only at positions corresponding to the variables $x_{i,1}, \dots, x_{i,n}$ where $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,n})$.
- $\mathbf{M}_{i,j}$ has nonzero columns only at positions corresponding to the cross-terms between $x_{i,1}, \dots, x_{i,n}$ and $x_{j,1}, \dots, x_{j,n}$.

Define the matrix

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{1,1} \\ \mathbf{M}_{2,2} \\ \vdots \\ \mathbf{M}_{k,k} \\ \mathbf{M}_{1,2} \\ \mathbf{M}_{1,3} \\ \vdots \\ \mathbf{M}_{k-1,k} \end{bmatrix} \in \mathbb{F}_q^{L(m-a) \times N}.$$

From the properties of the matrices $\mathbf{M}_{i,j}$ ($1 \leq i \leq j \leq k$), it follows that the matrix $\mathbf{M}$ has full row rank $L(m-a)$.

The Macaulay matrix representing $\mathcal{P}^*$ can then be written as

$$\mathbf{M}_{\mathcal{P}^*} = \mathbf{E}^{(R)} \cdot \mathbf{M},$$

where $\mathbf{E}^{(R)} \in \mathbb{F}_q^{m \times L(m-a)}$ is the horizontal concatenation of all emulsifier matrices:

$$\mathbf{E}^{(R)} = (\mathbf{E}_{1,1}^{(R)} \parallel \mathbf{E}_{2,2}^{(R)} \parallel \dots \parallel \mathbf{E}_{1,2}^{(R)} \parallel \dots \parallel \mathbf{E}_{k-1,k}^{(R)}).$$

Since $\mathbf{M}$ has full row rank, linear algebra implies that

$$\text{rank}(\mathbf{M}_{\mathcal{P}^*}) = \text{rank}(\mathbf{E}^{(R)}).$$

Therefore, it suffices to analyze the rank of $\mathbf{E}^{(R)}$.

Case $a = 0$. In this case,

$$\mathbf{E}^{(R)} = (\mathbf{A}_{1,1}^{(R)} \parallel \mathbf{A}_{2,2}^{(R)} \parallel \cdots \parallel \mathbf{A}_{k-1,k}^{(R)})$$

is a matrix of size $m \times Lm$. Since the matrices $\mathbf{A}_{i,j}^{(R)}$ are chosen corresponding to basis elements of $\mathbb{F}_{q^{m-a}}$, they are invertible. This implies that $\mathbf{E}^{(R)}$ contains at least one nonzero $m \times m$ minor. Hence $\mathbf{E}^{(R)}$ always has full rank m . This corresponds to the original MAYO case.

Case $a > 0$. Write

$$\mathbf{E}^{(R)} = \begin{pmatrix} \mathbf{E}_{\text{top}}^{(R)} \\ \mathbf{E}_{\text{bottom}}^{(R)} \end{pmatrix},$$

where $\mathbf{E}_{\text{top}}^{(R)} \in \mathbb{F}_q^{(m-a) \times L(m-a)}$ contains the first $m - a$ rows generated from the matrices $\mathbf{A}_{i,j}^{(R)}$, and $\mathbf{E}_{\text{bottom}}^{(R)} \in \mathbb{F}_q^{a \times L(m-a)}$ contains the remaining rows generated from the matrices $\mathbf{B}_{i,j}^{(R)}$.

By the same argument as in the case $a = 0$, the matrix $\mathbf{E}_{\text{top}}^{(R)}$ has full row rank $m - a$.

Let $V \subset \mathbb{F}_q^{L(m-a)}$ be the row space of $\mathbf{E}_{\text{top}}^{(R)}$, so $\dim(V) = m - a$. For $\mathbf{E}^{(R)}$ to have rank m , the a rows of $\mathbf{E}_{\text{bottom}}^{(R)}$ must lie outside V and must be linearly independent.

Since these rows are uniformly random and independent vectors in $\mathbb{F}_q^{L(m-a)}$ and independent of $\mathbf{E}_{\text{top}}^{(R)}$, the probability that the first row is not in V is

$$1 - q^{(m-a)-L(m-a)}.$$

Conditioned on this event, the probability that the second row lies outside the space spanned by V and the first row is

$$1 - q^{(m-a+1)-L(m-a)},$$

and so on.

Multiplying these probabilities together yields

$$\Pr[\text{rank}(\mathbf{E}^{(R)}) = m] = \prod_{i=0}^{a-1} \left(1 - q^{(m-a+i)-L(m-a)}\right).$$

□

Since $L(m - a) \gg m$, from Lemma 2 the probability that $\mathcal{P}^*$ produces a system whose rank is less than m is approximately $q^{m-L(m-a)}$. Therefore, as long as the value of a is not too large, this probability is negligible and can safely be ignored.

For the parameter set $(n, m, o, k, q, a) = (70, 64, 8, 8, 64, 12)$ as specified in the next section, the resulting probability is approximately 2^{-10848} .

Corollary 1. *Let c be an integer such that $c(m - a) > m$ and $c < L$. Then, the probability that the matrix $\mathbf{E}^{(R)}$ is full rank, when c matrices $\mathbf{B}_{i,j}$ are chosen arbitrarily, is lower bounded by*

$$\prod_{i=0}^{a-1} \left(1 - q^{(m-a+i)-c(m-a)}\right).$$

Proof. The proof follows the same line of reasoning as in Lemma 2, but restricted to a selection of c blocks while ignoring the remaining $L - c$ blocks.

In this case, $\mathbf{E}^{(R)}$ has full rank if the submatrix formed by these c blocks achieves rank m . This submatrix has dimension $m \times cm$, with its first $m - a$ rows already forming a full-rank submatrix of rank $m - a$.

Applying the argument from Lemma 2 to the remaining a rows, the probability that the submatrix is full rank is given by

$$\prod_{i=0}^{a-1} \left(1 - q^{(m-a+i)-c(m-a)}\right).$$

□

For the parameter set $(n, m, o, k, q, a) = (70, 64, 8, 8, 64, 12)$, as proposed in the next section, choosing $c = k$ yields a failure probability (i.e., the probability of not being full rank) of approximately 2^{-2118} . Although this probability is significantly larger than in the previous setting, it remains negligible and can therefore be safely ignored.

From the above corollary, we can freely choose the remaining $L - c$ matrices $\mathbf{B}_{i,j}$ without affecting the fact that the mapping $\mathcal{P}^*$ produces m linearly independent equations. This observation allows us to accelerate the signing and verification algorithms by selecting these matrices with a specific structure.

Proposal 1: Zero matrices. A straightforward approach is to set all $L - c$ matrices $\mathbf{B}_{i,j}$ to zero, which yields the maximum speed-up. However, this choice makes the last a equations of $\mathcal{P}^*$ highly sparse, which may introduce structural weaknesses that could be exploited by an attacker.

Proposal 2: Scalar block matrices. To mitigate this issue, we consider a simple structured alternative. Specifically, we partition each matrix $\mathbf{B}_{i,j}^{(R)}$ into sub-blocks of size $a \times a$, and choose each sub-block to be a scalar multiple of the identity matrix, i.e., of the form $\gamma \mathbf{I}_a$ for some $\gamma \in \mathbb{F}_q$. This construction significantly reduces sparsity while maintaining very efficient computation, since matrix-vector multiplication reduces to scalar multiplications.

Nevertheless, this approach only partially mitigates the sparsity issue and remains structurally restrictive, providing limited mixing between coordinates.

Proposal 3: Block-circulant matrices. To further enhance diffusion, we generalize this construction by replacing the scalar identity blocks with circulant matrices.

More precisely, each matrix $\mathbf{B}_{i,j}$ is partitioned into sub-blocks of size $a \times a$, where each sub-block is a circulant matrix defined by its first row, and the remaining rows are obtained by successive cyclic shifts. If the dimensions of $\mathbf{B}_{i,j}$ are not divisible by a , the entries that do not belong to any complete $a \times a$ sub-block are set to zero.

Although circulant blocks are slightly more expensive to compute than scalar identity blocks, they still allow for efficient implementations, while providing significantly better mixing properties. Based on our observations with existing attacks, this block-circulant design does not appear to weaken the security of the scheme and offers a favorable trade-off between efficiency and security.

7 MAYO ^-_L

7.1 Description of MAYO ^-_L

In this section, we propose a variant of MAYO denoted MAYO ^-_L by combining the rectangular improvement and the lifting technique introduced in the previous sections in order to obtain a scheme with smaller public key and signature sizes. Specifically, we select emulsifier matrices

$\mathbf{E}_{i,j}^{(RL)} \in \mathbb{F}_{q^r}^{m \times (m-a)}$ whose first $m-a$ rows form a matrix $\mathbf{A}_{i,j}^{(RL)}$ corresponding to a basis element of $\mathbb{F}_{(q^r)^{m-a}}$, while the remaining a rows form a matrix $\mathbf{B}_{i,j}^{(RL)}$.

The matrices $\mathbf{B}_{i,j}^{(RL)}$ are constructed as follows: for $i = j$, they are sampled uniformly at random over $\mathbb{F}_{q^r}^{a \times (m-a)}$; for $i \neq j$, they are structured as block-circulant matrices composed of circulant sub-blocks, as described in the previous section. More precisely, each circulant sub-block is fully determined by its first row, which is sampled uniformly at random over $\mathbb{F}_{q^r}$. The remaining rows of the sub-block are obtained by successive cyclic shifts of this first row. This construction allows for a compact representation and efficient computation while maintaining sufficient randomness.

Let $\mathcal{P}_1^*$ and $\mathcal{P}_2^*$ denote the first $m-a$ equations and the last a equations of $\mathcal{P}^*$, respectively. Based on this construction, we propose the MAYO_L^- scheme as illustrated in Figure 2.

For the key generation algorithm, MAYO_L^- also adopts the key generation technique of Petzoldt, as in the original MAYO scheme. In particular, the matrix representation of the map $\mathcal{P}$ now contains only $m-a$ equations and can be written as

$$\mathbf{P}_i = \begin{pmatrix} \mathbf{P}_i^{(1)} & \mathbf{P}_i^{(2)} \\ 0 & \mathbf{P}_i^{(3)} \end{pmatrix}, \quad (1 \leq i \leq m-a).$$

Here, $\mathbf{P}^{(1)} = \{\mathbf{P}_i^{(1)}\}_{i \in [m-a]}$ and $\mathbf{P}^{(2)} = \{\mathbf{P}_i^{(2)}\}_{i \in [m-a]}$ are still generated from the public seed, while $\mathbf{P}^{(3)} = \{\mathbf{P}_i^{(3)}\}_{i \in [m-a]}$ is computed so that the map vanishes on the oil space. Therefore, the main differences between MAYO_L^- and the original MAYO scheme lie in the signing and verification algorithms.

With the rectangular improvement, the algorithm introduces additional steps: Steps 3 and 6 in the signing algorithm, as well as Steps 2 and 5 in the verification algorithm. When the parameter $a = 0$, these steps are omitted and the scheme reduces to the original MAYO algorithm.

Regarding the lifting technique, the algorithm samples the vinegar variables $\mathbf{v}_i$ in the extension field $\mathbb{F}_{q^r}$ and solves the system $\mathcal{P}^*(\mathbf{v} + \mathbf{o}) = \mathbf{t}$ over the extension field $\mathbb{F}_{q^r}$.

Furthermore, from our experiments, the failure probability at Step 8 in the MAYO_L^- scheme is approximately $q^{r(m-ko-1)}$.

Finally, we note that the matrices $\mathbf{B}_{i,j}^{(RL)}$ can be chosen deterministically rather than sampled at random. In particular, one can construct all $\mathbf{B}_{i,j}^{(RL)}$ using block-circulant structures, as described previously, in such a way that the resulting matrix $\mathbf{E}^{(RL)}$ always has full rank. This eliminates the need for random generation while preserving the full-rank property of $\mathcal{P}^*$. Moreover, fixing these matrices improves the efficiency of both signing and verification, as they no longer need to be derived from a public seed.

On the other hand, if one wishes to prioritize security over efficiency, it is also possible to sample all matrices $\mathbf{B}_{i,j}^{(RL)}$ uniformly at random instead of imposing any specific structure. This avoids potential structural weaknesses at the cost of increased computational overhead.

7.2 Security Proof

In this section, we provide a security proof for MAYO_L^- . Similar to MAYO, the proposed scheme relies on the hardness of the UOV problem and on a new computational assumption closely related to the Multi-Target Whipped MQ problem, which we call the Multi-Target Whipped MQ with Rectangular and Lifting (MTWMQ $_L^-$) problem.

Therefore, the security reduction presented in this section is intended to provide cryptanalysts with the necessary framework for analyzing and potentially attacking the proposed scheme.

KeyGen():

- 1: $\mathbf{O} \leftarrow \mathbb{F}_q^{o \times (n-o)}$
- 2: seed $\leftarrow \{0, 1\}^{\text{pk_seed_length}}$
- 3: **for** $i = 1$ to $m - a$ **do**
- 4: $\mathbf{P}_i^{(1)} \leftarrow \text{Expand}(\text{seed} \parallel \text{P1} \parallel i)$ $\triangleright$ upper triangular $(n - o) \times (n - o)$ matrix
- 5: $\mathbf{P}_i^{(2)} \leftarrow \text{Expand}(\text{seed} \parallel \text{P2} \parallel i)$ $\triangleright o \times (n - o)$ matrix
- 6: $\mathbf{P}_i^{(3)} \leftarrow \text{Upper}(-\mathbf{O}\mathbf{P}_i^{(1)}\mathbf{O}^T - \mathbf{O}\mathbf{P}_i^{(2)})$
- 7: **end for**
- 8: **return** (pk, sk) = ((seed, $\{\mathbf{P}_i^{(3)}\}_{i \in [m-a]}$), (seed, $\mathbf{O}$))

Sign(M, sk):

- 1: (seed, $\mathbf{O}$) $\leftarrow$ sk
- 2: salt $\leftarrow \{0, 1\}^{\text{salt_length}}$
- 3: $\mathbf{B}^{(RL)} \leftarrow \text{GenerateB}(m, a, k, q, r)$ $\triangleright$ list containing $k(k+1)/2$ $a \times (m-a)$ -matrices
- 4: $\mathbf{t} \leftarrow \text{Hash}(M \parallel \text{salt}) \in \mathbb{F}_{q^r}^m$
- 5: $\mathcal{P}_1^*(\mathbf{x}_1, \dots, \mathbf{x}_k) \leftarrow \sum_{i=1}^k \mathbf{A}_{i,i}^{(RL)} \mathcal{P}(\mathbf{x}_i) + \sum_{1 \leq i < j \leq k} \mathbf{A}_{i,j}^{(RL)} \mathcal{P}'(\mathbf{x}_i, \mathbf{x}_j)$
- 6: $\mathcal{P}_2^*(\mathbf{x}_1, \dots, \mathbf{x}_k) \leftarrow \sum_{i=1}^k \mathbf{B}_{i,i}^{(RL)} \mathcal{P}(\mathbf{x}_i) + \sum_{1 \leq i < j \leq k} \mathbf{B}_{i,j}^{(RL)} \mathcal{P}'(\mathbf{x}_i, \mathbf{x}_j)$
- 7: $\mathbf{v}_i \leftarrow \mathbb{F}_{q^r}^{n-o} \times \{0\}^o$
- 8: **if** $\mathcal{P}^*(\mathbf{v}_1 + \mathbf{o}_1, \dots, \mathbf{v}_k + \mathbf{o}_k)$ does not have full rank **then**
- 9: go to Step 7
- 10: **end if**
- 11: Solve $\mathcal{P}^*(\mathbf{v}_1 + \mathbf{o}_1, \dots, \mathbf{v}_k + \mathbf{o}_k) = \mathbf{t}$ for $\mathbf{o}_1, \dots, \mathbf{o}_k \in \text{RowSpace}((\mathbf{O} \mathbf{I}_o))$
- 12: **return** $\sigma = (\text{salt}, \{\mathbf{s}_i = \mathbf{v}_i + \mathbf{o}_i\}_{i \in [k]})$

Verify(M, pk, σ):

- 1: (salt, $\{\mathbf{s}_i\}_{i \in [k]}$) $\leftarrow$ σ
- 2: $\mathbf{B}^{(RL)} \leftarrow \text{GenerateB}(m, a, k, q, r)$
- 3: $\mathbf{t} \leftarrow \text{Hash}(M \parallel \text{salt}) \in \mathbb{F}_{q^r}^m$
- 4: $\mathbf{t}'_1 \leftarrow \sum_{i=1}^k \mathbf{A}_{i,i}^{(RL)} \mathcal{P}(\mathbf{s}_i) + \sum_{1 \leq i < j \leq k} \mathbf{A}_{i,j}^{(RL)} \mathcal{P}'(\mathbf{s}_i, \mathbf{s}_j)$
- 5: $\mathbf{t}'_2 \leftarrow \sum_{i=1}^k \mathbf{B}_{i,i}^{(RL)} \mathcal{P}(\mathbf{s}_i) + \sum_{1 \leq i < j \leq k} \mathbf{B}_{i,j}^{(RL)} \mathcal{P}'(\mathbf{s}_i, \mathbf{s}_j)$
- 6: **return** accept if $\mathbf{t} = (\mathbf{t}'_1, \mathbf{t}'_2)$ and reject otherwise

Figure 2: Key generation, signing and verification of MAYO $_L^-$

Definition 1 (UOV problem). Let $\mathbf{O} \in \mathbb{F}_q^{(n-o) \times o}$ and let $\mathbf{MQ}_{n,m,q}(\mathbf{O})$ denote the set of multivariate quadratic maps $\mathcal{P} \in \mathbf{MQ}_{n,m,q}$ that vanish on the row space of $(\mathbf{O}^\top \mid \mathbf{I}_o)$.

The UOV problem is the problem of distinguishing a random multivariate quadratic map $\mathcal{P} \in \mathbf{MQ}_{n,m,q}$ from a random element of $\mathbf{MQ}_{n,m,q}(\mathbf{O})$, where $\mathbf{O} \leftarrow \mathbb{F}_q^{(n-o) \times o}$ is chosen uniformly at random.

Let $\mathcal{A}$ be a distinguisher. The distinguishing advantage of $\mathcal{A}$ is defined as

$$\text{Adv}_{n,m,o,q}^{\text{UOV}}(\mathcal{A}) = \left| \Pr[\mathcal{A}(\mathcal{P}) = 1 \mid \mathcal{P} \leftarrow \mathbf{MQ}_{n,m,q}] - \Pr\left[\mathcal{A}(\mathcal{P}) = 1 \mid \begin{array}{l} \mathbf{O} \leftarrow \mathbb{F}_q^{(n-o) \times o} \\ \mathcal{P} \leftarrow \mathbf{MQ}_{n,m,q}(\mathbf{O}) \end{array} \right] \right|.$$

Definition 2 (MTWMQ ^-_L problem). Let $\{\mathbf{E}_{i,j}^{(RL)}\}_{1 \leq i \leq j \leq k}$ be matrices in $\mathbb{F}_{q^r}^{m \times (m-a)}$ defined as in MAYO ^-_L . Given a random quadratic map $\mathcal{P} \in \mathbf{MQ}_{n,m-a,q}$ and access to an unbounded number of random targets $\mathbf{t}_i \leftarrow \mathbb{F}_{q^r}^m$ for $i \in \mathbb{N}$, the multi-target whipped MQ with rectangular and lifting problem asks to compute $(I, \mathbf{s}_1, \dots, \mathbf{s}_k)$ such that

$$\sum_{i=1}^k \mathbf{E}_{i,i}^{(RL)} \mathcal{P}(\mathbf{s}_i) + \sum_{1 \leq i < j \leq k} \mathbf{E}_{i,j}^{(RL)} \mathcal{P}'(\mathbf{s}_i, \mathbf{s}_j) = \mathbf{t}_I.$$

Let $\mathcal{A}$ be an adversary. The success probability of $\mathcal{A}$ in solving the MTWMQ ^-_L problem is defined as

$$\text{Adv}_{n,m,k,q,a,r}^{\text{MT-WMQ}^-_L}(\mathcal{A}) = \Pr \left[\sum_{i=1}^k \mathbf{E}_{i,i}^{(RL)} \mathcal{P}(\mathbf{s}_i) + \sum_{i < j} \mathbf{E}_{i,j}^{(RL)} \mathcal{P}'(\mathbf{s}_i, \mathbf{s}_j) = \mathbf{t}_I \mid \begin{array}{l} \mathcal{P} \leftarrow \mathbf{MQ}_{n,m-a,q} \\ \{\mathbf{t}_i\} \leftarrow \mathbb{F}_{q^r}^{m \times \mathbb{N}} \\ (I, \mathbf{s}_1, \dots, \mathbf{s}_k) \leftarrow \mathcal{A}^{\mathbf{t}_i}(\mathcal{P}) \end{array} \right].$$

Theorem 1. Let $\mathcal{A}$ be an EUF-CMA adversary running in time T against the MAYO ^-_L signature scheme in the random oracle model. Assume that $\mathcal{A}$ makes at most Q_s signing queries and Q_h random oracle queries.

Let

$$\tau = q^{r(m-ko-1)}$$

be the failure probability. If $Q_s \tau < 1$, then there exist adversaries $\mathcal{B}^{\mathcal{A}}$ and $\mathcal{B}'^{\mathcal{A}}$ against the problems UOV $_{n,m-a,o,q}$ and MTWMQ ^-_L $_{n,m,k,q,a,r}$ respectively, running in time

$$T + (Q_s + Q_h + 1) \cdot \text{poly}(n, m, k, q, a, r),$$

such that

$$\begin{aligned} \text{Adv}_{n,m,o,k,q,a,r}^{\text{EUF-CMA}}(\mathcal{A}) &\leq \left(\text{Adv}_{n,m-a,o,q}^{\text{UOV}}(\mathcal{B}^{\mathcal{A}}) + \text{Adv}_{n,m,k,q,a,r}^{\text{MTWMQ}^-_L}(\mathcal{B}'^{\mathcal{A}}) + q^{-rm} \right) (1 - Q_s \tau)^{-1} \\ &\quad + (Q_h + Q_s) Q_s 2^{-\text{salt_length}}. \end{aligned}$$

The proof follows the same arguments as in MAYO [15]; for details we refer to Appendix B.

7.3 Security Analysis of New Variant

In this section, we analyze the known attacks against the proposed scheme.

7.3.1 Direct Attack 1

Effect of the rectangular improvement. Corollary 1 shows that the map $\mathcal{P}^*$ produces a quadratic system with kn variables and m equations over $\mathbb{F}_q$ with high probability. Hence, the cost of the attack remains the same as for the original MAYO scheme, namely

$$MQ(kn, m, q)$$

Effect of the lifting technique. Since the coefficients of the equation $\mathcal{P}^*(\mathbf{s}) = \mathbf{t}$ lie in $\mathbb{F}_{q^r}$, an attacker must solve an MQ system with kn variables and m equations over $\mathbb{F}_{q^r}$. Therefore the cost of this attack is

$$MQ(kn, m, q^r)$$

MAYO_L⁻. Since Corollary 1 shows that, with high probability, $\mathcal{P}^*$ forms a system of m equations in kn variables, and since the attack complexity is also influenced by the lifting technique, it can be estimated as

$$MQ(kn, m, q^r)$$

7.3.2 Direct Attack 2

Effect of the rectangular improvement. Since the matrix $\mathbf{E}_\alpha$ in Equation (2) now has rank $m - a$, we can apply Gaussian elimination to Equation (2) in order to eliminate the quadratic part in the last a equations. Substituting the resulting a linear relations into the remaining equations yields a quadratic system with $n - a$ variables and $m - a$ equations. The cost of the attack in this case is therefore

$$MQ(n - a, m - a, q)$$

Effect of the lifting technique. An attacker may perform the change of variables $\mathbf{x}_i = \alpha_i \mathbf{x}_1$ for all $2 \leq i \leq k$ in Equation (1), which leads to

$$\mathbf{E}_\alpha \mathcal{P}(\mathbf{x}_1) = \mathbf{t}. \tag{3}$$

Multiplying both sides by $\mathbf{E}_\alpha^{-1}$ yields exactly the lifted MQ system consisting of n variables and m equations. Hence, an attacker may attempt to apply lifted-system solving attacks to this system. However, since the parameters m and n are chosen to be close to each other, the success probability of such attacks is negligible. The cost of the attack therefore remains

$$MQ(n, m, q^r)$$

MAYO_L⁻. Since both modifications are applied, the attacker first performs the above change of variables to obtain Equation (3) and then exploits the fact that $\mathbf{E}_\alpha$ has rank $m - a$. The attacker must therefore solve a quadratic system with $n - a$ variables and $m - a$ equations over $\mathbb{F}_{q^r}$. Although the coefficients of this system lie in $\mathbb{F}_q$, which in principle allows the application of attacks such as SDA, NSDA, and NSDA+, these attacks cannot be applied in our parameter setting since m and n are chosen to be close. Consequently, the cost of the attack is

$$MQ(n - a, m - a, q^r)$$

7.3.3 Direct Attack 3

Effect of the rectangular improvement. Similarly to Direct Attack 2, we perform the change of variables $\mathbf{x}_i = \alpha_i \mathbf{x}_1$ for all $2 \leq i \leq k$, which yields Equation (3). If $\mathbf{t}$ lies in the image space $\text{Im}(\mathbf{E}_\alpha)$, the problem reduces to solving a quadratic multivariate system consisting of $m - a$ equations in n variables.

On the other hand, the probability that $\mathbf{t}$ belongs to $\text{Im}(\mathbf{E}_\alpha)$ is q^{-a} . Therefore, by trying q^a different values of the *salt*, an attacker can expect to obtain a suitable vector $\mathbf{t}$. The cost of the attack is thus

$$q^a \lambda + MQ(n, m - a, q)$$

Effect of the lifting technique. Since there is no rank reduction in the matrix $\mathbf{E}_\alpha$, this attack coincides with Direct Attack 2. Therefore, the cost remains

$$MQ(n, m, q^r)$$

MAYO $_L^-$. In the case of MAYO $_L^-$, the attack complexity is mainly influenced by the rectangular improvement, which introduces a rank reduction in the matrix $\mathbf{E}_\alpha$. Consequently, the attacker only needs to solve a quadratic multivariate system with $m - a$ equations and n variables. The resulting attack cost is

$$q^{ar} \lambda + MQ(n, m - a, q^r)$$

Since this attack results in the largest gap between the number of variables and the number of equations, we will estimate the security level in the following section based on the parameters of this system.

7.3.4 Collision Attack

Effect of the rectangular improvement. Similarly to Direct Attacks 2 and 3, the attacker can exploit the reduced rank of the matrix $\mathbf{E}_\alpha$ to obtain an MQ system with fewer equations and then perform a collision search on this system. More precisely, the attacker performs the collision search based on Equation (3). Consequently, the cost of the attack is

$$ColAtk(m - a, q)$$

Effect of the lifting technique. Since there is no rank reduction and the resulting system is defined over the field $\mathbb{F}_{q^r}$, the cost of the attack becomes

$$ColAtk(m, q^r)$$

MAYO $_L^-$. When both the rectangular improvement and the lifting technique are applied, the attacker must perform a collision search for a quadratic multivariate system consisting of $m - a$ equations over $\mathbb{F}_{q^r}$. The resulting attack cost in this case is therefore

$$ColAtk(m - a, q^r)$$

7.3.5 Key-Recovery Attack

Effect of the rectangular improvement. The attacker must perform key recovery on an instance of $\text{UOV}(n, m - a, o, q)$.

Effect of the lifting technique. The key-recovery attack is the same as in the original MAYO scheme, and thus the attacker faces an instance of $\text{UOV}(n, m, o, q)$.

MAYO $_L^-$. The attacker must perform key recovery on an instance of $\text{UOV}(n, m - a, o, q)$.

7.4 Parameter Choices

In this section, we present the proposed parameter sets for MAYO_L^- and estimate their security against the known attacks described above. For a fair comparison with MAYO, we adopt parameter sets that target the same security levels as MAYO, namely 156, 224, and 296 bits. In addition, the parameters are chosen such that the Rectangular MinRank Attack does not apply, as in MAYO, i.e., $n - (m - a) \geq o$.

Table 2 reports the success probabilities of the SDA, NSDA, and NSDA+ attacks against MAYO_L^- . Since the gap between the number of variables and the number of equations in Direct Attack 3 is larger than in Direct Attack 2, we only report the success probabilities of SDA, NSDA, and NSDA+ corresponding to Direct Attack 3 in Table 2.

Table 3 summarizes the complexity of various attacks, including the Collision Attack (Col), the Hashimoto algorithm combined with WXL for Direct Attacks 1, 2, and 3, the Kipnis–Shamir attack (KS), the Reconciliation Attack (Rec), and the Intersection Attack (Int). Note that the Wedge Attack does not satisfy the condition required to find a suitable o' , and therefore cannot be applied to attack the parameter set we propose.

Let $\mathbf{esk}$ and $\mathbf{epk}$ denote the expanded secret key and expanded public key, respectively. The size of $\mathbf{esk}$ consists of the size of the matrix $\mathbf{O}$, the matrices $\{\mathbf{P}_i^{(1)}\}_{i \in [m-a]}$, and the matrices $\mathbf{L}_i = (\mathbf{P}_i^{(1)} + \mathbf{P}_i^{(1)T})\mathbf{O} + \mathbf{P}_i^{(2)}$. The size of $\mathbf{epk}$ includes the matrices $\mathbf{P}^{(1)}$, $\mathbf{P}^{(2)}$, and $\mathbf{P}^{(3)}$.

Table 4 compares the public key and signature sizes of MAYO_L^- with those of the original MAYO scheme and the standardized Dilithium scheme [45]. For the parameter sets targeting security levels I, III, and V, we also use a public seed of length 16 bytes and salts of length 24, 32, and 40 bytes, respectively, following the same choices as in MAYO.

Table 2: Success probability of SDA, NSDA, and NSDA+

Level	Parameters	SDA	NSDA	NSDA+
I	(70, 64, 8, 8, 8, 2, 12)	2^{-102}	2^{-102}	2^{-102}
	(62, 56, 8, 7, 16, 2, 8)	2^{-136}	2^{-136}	2^{-136}
III	(105, 92, 11, 9, 8, 2, 13)	2^{-159}	2^{-159}	2^{-159}
	(102, 96, 8, 12, 8, 2, 17)	2^{-168}	2^{-168}	2^{-168}
	(91, 87, 8, 11, 16, 2, 13)	2^{-228}	2^{-228}	2^{-228}
V	(139, 128, 10, 13, 8, 2, 19)	2^{-237}	2^{-237}	2^{-237}
	(124, 116, 10, 12, 16, 2, 14)	2^{-320}	2^{-320}	2^{-320}

8 Applicability to SNOVA

In this section, we show that the improvement to the emulsifier matrix introduced in the previous section can also be applied to SNOVA. Since SNOVA is based on a UOV variant, its security evaluation is currently less well understood due to the continuous emergence of new attacks. Therefore, in this section we only analyze the applicability of the proposed improvements rather than providing a full security assessment.

Table 3: Cost estimate for each attack of MAYO_L^-

SL	(n, m, o, k, q, r, a)	$\log_2(\#\text{gates})$ in classical (upper) / quantum (lower)						
		KS	Rec	Int	Direct 1	Direct 2	Direct 3	Col
I	(70, 64, 8, 8, 8, 2, 12)	180	157	239	161	157	157	167
		101	102	165	117	129	129	167
	(62, 56, 8, 7, 16, 2, 8)	202	156	246	158	158	158	203
		112	111	165	123	134	134	203
III	(105, 92, 11, 9, 8, 2, 13)	269	225	348	226	224	224	249
		146	147	236	166	182	182	249
	(102, 96, 8, 12, 8, 2, 17)	278	225	360	226	224	224	249
		151	147	240	164	182	182	249
	(91, 87, 8, 11, 16, 2, 13)	319	225	385	224	224	224	307
		171	159	246	178	191	191	307
V	(139, 128, 10, 13, 8, 2, 19)	378	297	485	301	296	296	339
		201	193	317	221	239	239	339
	(124, 116, 10, 12, 16, 2, 14)	437	297	517	296	296	296	419
		230	208	327	241	250	250	419

Table 4: Key and signature sizes for MAYO, MAYO_L^- and Dilithium across security levels. All sizes are reported in bytes (B) or kilobytes (KB).

Level	Scheme	Parameters	pk (B)	sig (B)	esk (KB)	epk (KB)
I	MAYO	$(n, m, o, k, q) = (86, 78, 8, 10, 16)$	1420	454	142	143
	MAYO_L^-	$(n, m, o, k, q, r, a) = (70, 64, 8, 8, 8, 2, 12)$	718	444	47	48
	MAYO_L^-	$(n, m, o, k, q, r, a) = (62, 56, 8, 7, 16, 2, 8)$	876	458	46	46
	Dilithium	ML-DSA-44	1312	2420	–	–
III	MAYO	$(n, m, o, k, q) = (118, 108, 10, 11, 16)$	2986	681	368	371
	MAYO_L^-	$(n, m, o, k, q, r, a) = (105, 92, 11, 9, 8, 2, 13)$	1972	741	160	161
	MAYO_L^-	$(n, m, o, k, q, r, a) = (102, 96, 8, 12, 8, 2, 17)$	1083	950	152	152
	MAYO_L^-	$(n, m, o, k, q, r, a) = (91, 87, 8, 11, 16, 2, 13)$	1344	1033	151	152
	Dilithium	ML-DSA-65	1952	3309	–	–
V	MAYO	$(n, m, o, k, q) = (154, 142, 12, 12, 16)$	5554	964	823	828
	MAYO_L^-	$(n, m, o, k, q, r, a) = (139, 128, 10, 13, 8, 2, 19)$	2265	1396	387	389
	MAYO_L^-	$(n, m, o, k, q, r, a) = (124, 116, 10, 12, 16, 2, 14)$	2817	1528	384	386
	Dilithium	ML-DSA-87	2592	4627	–	–

8.1 Description of SNOVA

SNOVA is a UOV-based digital signature scheme proposed by Wang et al. [16]. Let $\mathcal{R} = \text{Mat}_{\ell \times \ell}(\mathbb{F}_q)$ denote the non-commutative ring of $\ell \times \ell$ matrices, and let $\mathbb{F}_q[\mathbf{S}] \subset \mathcal{R}$ be the subring generated by a symmetric $\ell \times \ell$ matrix $\mathbf{S}$ whose characteristic polynomial is irreducible.

The central map of SNOVA is described by the quadratic map $\tilde{\mathcal{F}} = (\tilde{f}_1, \dots, \tilde{f}_m) : \mathbb{F}_q^{n\ell^2} \rightarrow \mathbb{F}_q^{m\ell^2}$. Let $i' = i + \alpha \bmod m$. Each map $\tilde{f}_i : \mathbb{F}_q^{n\ell^2} \rightarrow \mathbb{F}_q^{\ell^2}$ is defined as

$$\tilde{f}_i(\mathbf{X}) = \sum_{\alpha=0}^{\ell^2+\ell-1} \mathbf{A}_{i,\alpha} \cdot \mathbf{X}^T \cdot \Lambda_{\mathbf{Q}_{i,\alpha,1}}^{(n)} \cdot \tilde{\mathbf{F}}_{i'} \cdot \Lambda_{\mathbf{Q}_{i,\alpha,2}}^{(n)} \mathbf{X} \cdot \mathbf{B}_{i,\alpha} \quad (1 \leq i \leq m),$$

where $\mathbf{X} = (\mathbf{X}_1^T, \dots, \mathbf{X}_n^T)^T$ is an $\ell n \times \ell$ matrix containing the $\ell \times \ell$ matrix variables. The matrices $\mathbf{Q}_{i,\alpha,1}, \mathbf{Q}_{i,\alpha,2} \in \mathbb{F}_q[\mathbf{S}]$ and $\mathbf{A}_{i,\alpha}, \mathbf{B}_{i,\alpha} \in \mathbb{F}_q^{\ell \times \ell}$ are public parameters, while $\tilde{\mathbf{F}}_{i'} \in \mathbb{F}_q^{\ell n \times \ell n}$ are secret matrices containing an $m \times m$ zero submatrix, similarly to the matrix representation of the central map in UOV. The matrices $\mathbf{A}_{i,\alpha}, \mathbf{B}_{i,\alpha}, \mathbf{Q}_{i,\alpha,1}, \mathbf{Q}_{i,\alpha,2}$ strongly influence the security of the scheme and are often collectively referred to as the $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{Q}$ matrices.

Similarly, the public map of SNOVA is given by the quadratic map $\tilde{\mathcal{P}} = (\tilde{p}_1, \dots, \tilde{p}_m) : \mathbb{F}_q^{n\ell^2} \rightarrow \mathbb{F}_q^{m\ell^2}$. Let $i' = i + \alpha \bmod m$. Each map $\tilde{p}_i : \mathbb{F}_q^{n\ell^2} \rightarrow \mathbb{F}_q^{\ell^2}$ is defined as

$$\tilde{p}_i(\mathbf{X}) = \sum_{\alpha=0}^{\ell^2+\ell-1} \mathbf{A}_{i,\alpha} \cdot \mathbf{X}^T \cdot \Lambda_{\mathbf{Q}_{i,\alpha,1}}^{(n)} \cdot \tilde{\mathbf{P}}_{i'} \cdot \Lambda_{\mathbf{Q}_{i,\alpha,2}}^{(n)} \mathbf{X} \cdot \mathbf{B}_{i,\alpha} \quad (1 \leq i \leq m),$$

where $\tilde{\mathbf{P}}_i = \tilde{\mathbf{T}}^T \tilde{\mathbf{F}}_i \tilde{\mathbf{T}} \in \mathbb{F}_q^{\ell n \times \ell n}$ is public and $\tilde{\mathbf{T}} \in (\mathbb{F}_q[\mathbf{S}])^{n \times n}$ is a secret invertible matrix.

The signing procedure of SNOVA is similar to that of UOV. First, the variables $\mathbf{X}_1, \dots, \mathbf{X}_{n-m}$ are fixed at random in the central map, which yields a linear system over $\mathbb{F}_q$ in the remaining variables $\mathbf{X}_{n-m+1}, \dots, \mathbf{X}_n$. Since this system consists of $m\ell^2$ variables and $m\ell^2$ equations, it has a solution with high probability. The signature is then obtained by computing $\mathbf{Y} = \tilde{\mathbf{T}}^{-1}(\mathbf{X})$.

For verification, one checks whether $\tilde{\mathcal{P}}(\mathbf{Y}) = H(M \parallel \text{salt})$.

On the other hand, Beullens [29] showed that the public map of SNOVA can also be described using the whipping technique and can be written as

$$\tilde{\mathcal{P}}(\mathbf{U}) = \sum_{i=1}^{\ell} \sum_{j=1}^{\ell} \tilde{\mathbf{E}}_{i,j} \mathcal{G}(\mathbf{u}_i, \mathbf{u}_j),$$

where $\mathbf{U} = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{\ell})$ with each $\mathbf{u}_i$ being a vector of ℓn variables, $\tilde{\mathbf{E}}_{i,j} \in \mathbb{F}_q^{m\ell^2 \times m\ell^2}$ are the emulsifier matrices constructed from the matrices $\mathbf{A}, \mathbf{B}, \mathbf{Q}$, and $\mathcal{G} : \mathbb{F}_q^{n\ell} \times \mathbb{F}_q^{n\ell} \rightarrow \mathbb{F}_q^{m\ell^2}$ is the bilinear map defined by

$$\mathcal{G}(\mathbf{x}, \mathbf{y}) := (\mathcal{G}_1(\mathbf{x}, \mathbf{y}), \dots, \mathcal{G}_m(\mathbf{x}, \mathbf{y}))$$

with

$$\mathcal{G}_h(\mathbf{x}, \mathbf{y}) := \left(\mathbf{x}^T \Lambda_{\mathbf{S}^{i-1}}^{(n)} \tilde{\mathbf{P}}_h \Lambda_{\mathbf{S}^{j-1}}^{(n)} \mathbf{y} \right)_{i,j \in [\ell]}$$

8.2 Applying the Lifting Enhancement to SNOVA's Emulsifier Matrices

The lifting modification can be applied directly to SNOVA without altering its simple signing and verification procedures. More precisely, we choose the matrices $\mathbf{A}$ and $\mathbf{B}$ over the extension field $\mathbb{F}_{q^r}^{\ell \times \ell}$ while keeping the remaining matrices unchanged. As a result, the emulsifier matrices $\tilde{\mathbf{E}}_{i,j}$ constructed from $\mathbf{A}$ and $\mathbf{B}$ will contain entries in $\mathbb{F}_{q^r}$.

The signing procedure remains similar to that of SNOVA. Specifically, we first fix the variables $\mathbf{X}_1, \dots, \mathbf{X}_{n-m}$ at random in $\mathbb{F}_{q^r}^{\ell \times \ell}$ in the central map, which yields a linear system over $\mathbb{F}_{q^r}$. After solving this system and obtaining a solution $\mathbf{X}$, the signature is computed as

$$\mathbf{Y} = \tilde{\mathbf{T}}^{-1}(\mathbf{X}).$$

9 Conclusion

In this paper, we studied and improved the whipping technique for UOV-based signature schemes with the goal of reducing key and signature sizes while maintaining security against known attacks.

In particular, we proposed two general improvements. First, we proposed lifting the emulsifier matrices to an extension field while keeping the underlying UOV instance defined over the base field. This modification mitigates the main drawback of LUOV, namely that lifted-system solving attacks typically exploit systems where the number of variables and equations are unbalanced. As a result, attacks such as SDA, NSDA, and NSDA+ become ineffective. Furthermore, this lifting enhancement strengthens resistance against forgery attacks, making existing modification techniques such as the use of rectangular emulsifier matrices or setting certain emulsifier matrices to zero significantly more effective. Second, we enhanced rectangular emulsifier matrices originally introduced in prior work by incorporating a structured block design that accelerates signing and verification while preserving the full-rank property.

By combining these two improvements, we proposed the MAYO ^-_L variant. The resulting scheme achieves smaller key and signature sizes while maintaining the same targeted security levels. In particular, the expanded key size is reduced by approximately 80 KB, 200 KB, and 420 KB for security levels I, III, and V, respectively.

Regarding security proofs, we provided a reduction framework similar to that of MAYO: the EUF-CMA security is reduced to EUF-KOA, which is then related to the hardness of distinguishing UOV instances and the MT-WMQ ^-_L problem. In addition, the appendix discusses a possible proof approach in the QROM based on the INV/WPSF assumptions.

Finally, we discussed the applicability of our proposed lifting improvement to SNOVA. The lifting modification can be integrated seamlessly by choosing the matrices A and B over an extension field, demonstrating that this enhancement is not limited to MAYO but can also benefit other UOV-based signature schemes employing the whipping technique.

These results suggest new directions for designing more flexible UOV-based signature schemes using the whipping technique and motivate further research on optimal parameter choices and tighter security proofs, especially in the QROM setting, for variants employing structured emulsifier matrices.

References

- [1] Peter W Shor. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM review*, 41(2):303–332, 1999.
- [2] National Institute of Standards and Technology. Post-quantum cryptography (pqc) project. <https://csrc.nist.gov/projects/post-quantum-cryptography>, 2025. Accessed: 2026-02-14.
- [3] National Institute of Standards and Technology. Post-quantum cryptography: Additional digital signature schemes. <https://csrc.nist.gov/projects/pqc-dig-sig>, 2025. Accessed: 2026-02-14.
- [4] Michael R Garey, David S Johnson, et al. A guide to the theory of np-completeness. *Computers and intractability*, pages 37–79, 1990.
- [5] Jacques Patarin. The oil and vinegar signature scheme. In *Presented at the Dagstuhl Workshop on Cryptography September 1997*, 1997.
- [6] Jacques Patarin. Cryptanalysis of the matsumoto and imai public key scheme of eurocrypt88. In *Annual International Cryptology Conference*, pages 248–261. Springer, 1995.
- [7] Tsutomu Matsumoto and Hideki Imai. Public quadratic polynomial-tuples for efficient signature-verification and message-encryption. In *Workshop on the Theory and Application of Cryptographic Techniques*, pages 419–453. Springer, 1988.
- [8] Aviad Kipnis and Adi Shamir. Cryptanalysis of the oil and vinegar signature scheme. In *Annual international cryptology conference*, pages 257–266. Springer, 1998.
- [9] Aviad Kipnis, Jacques Patarin, and Louis Goubin. Unbalanced oil and vinegar signature schemes. In *International Conference on the Theory and Applications of Cryptographic Techniques*, pages 206–222. Springer, 1999.
- [10] Albrecht Petzoldt, Enrico Thomae, Stanislav Bulygin, and Christopher Wolf. Small public keys and fast verification for multivariate quadratic public key systems. In *International Workshop on Cryptographic Hardware and Embedded Systems*, pages 475–490. Springer, 2011.
- [11] Ward Beullens and Bart Preneel. Field lifting for smaller uov public keys. In *International Conference on Cryptology in India*, pages 227–246. Springer, 2017.
- [12] Jintai Ding, Joshua Deaton, Vishakha, and Bo-Yin Yang. The nested subset differential attack: a practical direct attack against luov which forges a signature within 210 minutes. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, pages 329–347. Springer, 2021.
- [13] Alan Szepieniec and Bart Preneel. Block-anti-circulant unbalanced oil and vinegar. In *International Conference on Selected Areas in Cryptography*, pages 574–588. Springer, 2019.
- [14] Hiroki Furue, Yasuhiko Ikematsu, Yutaro Kiyomura, and Tsuyoshi Takagi. A new variant of unbalanced oil and vinegar using quotient ring: Qr-uov. In *International Conference on the Theory and Application of Cryptology and Information Security*, pages 187–217. Springer, 2021.

- [15] Ward Beullens. Mayo: practical post-quantum signatures from oil-and-vinegar maps. In *International Conference on Selected Areas in Cryptography*, pages 355–376. Springer, 2021.
- [16] Lih-Chung Wang, Po-En Tseng, Yen-Liang Kuan, and Chun-Yen Chou. A simple noncommutative uov scheme. *Cryptology ePrint Archive*, 2022.
- [17] Peigen Li, Hao Guo, and Jintai Ding. Unified approach to UOV-like multivariate signature schemes. *Cryptology ePrint Archive*, Paper 2025/1778, 2025.
- [18] Yasuhiko Ikematsu. A survey on small public key signature schemes derived from uov signature scheme. In *Mathematical Foundations for Post-Quantum Cryptography: Crypto-Math CREST*, pages 313–332. Springer, 2025.
- [19] Inès Van De Weyer. Emulsifier map for mayo: A new multivariate quadratic signature scheme in post-quantum cryptography. Master’s thesis, KU Leuven, Leuven, Belgium, 2024.
- [20] Nicolas Courtois, Alexander Klimov, Jacques Patarin, and Adi Shamir. Efficient algorithms for solving overdefined systems of multivariate polynomial equations. In *International Conference on the Theory and Applications of Cryptographic Techniques*, pages 392–407. Springer, 2000.
- [21] Jean-Charles Faugere. A new efficient algorithm for computing gröbner bases (f4). *Journal of pure and applied algebra*, 139(1-3):61–88, 1999.
- [22] Jean Charles Faugere. A new efficient algorithm for computing gröbner bases without reduction to zero (f 5). In *Proceedings of the 2002 international symposium on Symbolic and algebraic computation*, pages 75–83, 2002.
- [23] Bo-Yin Yang, Owen Chia-Hsin Chen, Daniel J Bernstein, and Jiun-Ming Chen. Analysis of quad. In *International Workshop on Fast Software Encryption*, pages 290–308. Springer, 2007.
- [24] Lov K Grover. A fast quantum mechanical algorithm for database search. In *Proceedings of the twenty-eighth annual ACM symposium on Theory of computing*, pages 212–219, 1996.
- [25] Enrico Thomae and Christopher Wolf. Solving underdetermined systems of multivariate quadratic equations revisited. In *International workshop on public key cryptography*, pages 156–171. Springer, 2012.
- [26] Yasufumi Hashimoto. An improvement of algorithms to solve under-defined systems of multivariate quadratic equations. *Cryptology ePrint Archive*, 2021.
- [27] Jintai Ding, Zheng Zhang, Joshua Deaton, Kurt Schmidt, and F Vishakha. New attacks on lifted unbalanced oil vinegar. In *the 2nd NIST PQC Standardization Conference*, 2019.
- [28] Ward Beullens. Improved cryptanalysis of uov and rainbow. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, pages 348–373. Springer, 2021.
- [29] Ward Beullens. Improved cryptanalysis of snova. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, pages 277–293. Springer, 2025.
- [30] Charles Bouillaguet, Hsieh-Chung Chen, Chen-Mou Cheng, Tung Chou, Ruben Niederhagen, Adi Shamir, and Bo-Yin Yang. Fast exhaustive search for polynomial systems in. In *International Workshop on Cryptographic Hardware and Embedded Systems*, pages 203–218. Springer, 2010.

- [31] Samuel Jaques and John M Schanck. Quantum cryptanalysis in the ram model: Claw-finding attacks on sike. In *Annual International Cryptology Conference*, pages 32–61. Springer, 2019.
- [32] Pierre Pébereau. One vector to rule them all: Key recovery from one vector in uov schemes. In *International Conference on Post-Quantum Cryptography*, pages 92–108. Springer, 2024.
- [33] Jintai Ding, Bo-Yin Yang, Chia-Hsin Owen Chen, Ming-Shing Chen, and Chen-Mou Cheng. New differential-algebraic attacks and reparametrization of rainbow. In *International Conference on Applied Cryptography and Network Security*, pages 242–257. Springer, 2008.
- [34] Hiroki Furue and Yasuhiko Ikematsu. A new security analysis against mayo and qr-uov using rectangular minrank attack. In *International Workshop on Security*, pages 101–116. Springer, 2023.
- [35] Magali Bardet, Maxime Bros, Daniel Cabarcas, Philippe Gaborit, Ray Perlner, Daniel Smith-Tone, Jean-Pierre Tillich, and Javier Verbel. Improvements of algebraic attacks for solving the rank decoding and minrank problems. In *International Conference on the Theory and Application of Cryptology and Information Security*, pages 507–536. Springer, 2020.
- [36] Lars Ran. Wedges, oil, and vinegar—an analysis of uov in characteristic 2. *Cryptology ePrint Archive*, 2025.
- [37] Yi Jin, Yuansheng Pan, Xiaou He, Boru Gong, and Jintai Ding. Security analysis on uov families with odd characteristics: Using symmetric algebra. *Cryptology ePrint Archive*, 2025.
- [38] Werner H. Greub. *Multilinear Algebra*. Universitext. Springer, New York, NY, 1978.
- [39] Don Coppersmith. Solving homogeneous linear equations over $\text{GF}(2)$ via block wiedemann algorithm. *Mathematics of Computation*, 62(205):333–350, 1994.
- [40] Joshua Deaton. *A Cryptanalysis of Lifted Underdetermined Multivariate Cryptosystems*. PhD thesis, University of Cincinnati, 2022.
- [41] Itai Dinur. Cryptanalytic applications of the polynomial method for solving multivariate equation systems over $\text{GF}(2)$. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, pages 374–403. Springer, 2021.
- [42] Antoine Joux and Vanessa Vitse. A crossbred algorithm for solving boolean polynomial systems. *Cryptology ePrint Archive*, Paper 2017/372, 2017.
- [43] Andre Esser, Javier Verbel, Floyd Zveydinger, and Emanuele Bellini. Cryptographicestimators: a software library for cryptographic hardness estimation. *IACR Cryptol. ePrint Arch*, page 589, 2023.
- [44] David P. Wagner. A generalized birthday problem (extended abstract). In *Annual International Cryptology Conference*, 2002.
- [45] National Institute of Standards and Technology. Module-lattice-based digital signature standard. Federal Information Processing Standards Publication (FIPS) 204, August 2024. NIST, Gaithersburg, MD, USA.
- [46] Haruhisa Kosuge and Keita Xagawa. Probabilistic hash-and-sign with retry in the quantum random oracle model. In *IACR International Conference on Public-Key Cryptography*, pages 259–288. Springer, 2024.

- [47] Alex B Grilo, Kathrin Hövelmanns, Andreas Hülsing, and Christian Majenz. Tight adaptive reprogramming in the qrom. In *International Conference on the Theory and Application of Cryptology and Information Security*, pages 637–667. Springer, 2021.
- [48] Jelle Don, Serge Fehr, and Christian Majenz. The measure-and-reprogram technique 2.0: Multi-round fiat-shamir and more. In *Annual International Cryptology Conference*, pages 602–631. Springer, 2020.

Appendix A: MAYO_L

In this section, we propose parameter sets for the MAYO_L scheme to demonstrate the effectiveness of the lifting improvement. As in the Appendix A, the parameter sets are chosen to target security levels of 156, 224, and 296 bits. In addition, we maintain the condition $n - m \geq o$, as in the original MAYO scheme.

Table 5 presents the estimated complexities of known attacks against MAYO_L. These include the Kipnis–Shamir attack, the Reconciliation attack, the Intersection attack, Direct Attack 1, Direct Attack 2, Direct Attack 3, and the Collision attack.

Table 5: Cost estimate for each attack of MAYO _L								
SL	(n, m, o, k, q, r)	$\log_2(\#\text{gates})$ in classical (upper) / quantum (lower)						
		KS	Rec	Int	Direct 1	Direct 2	Direct 3	Col
I	(75, 63, 9, 7, 8, 2)	190	158	241	161	184	184	201
		106	108	167	118	151	151	201
	(76, 64, 8, 8, 8, 2)	199	162	254	159	187	187	204
		110	110	173	113	151	151	204
III	(110, 93, 11, 9, 8, 2)	284	224	360	228	257	257	291
		154	152	240	166	208	208	291
	(110, 96, 10, 10, 8, 2)	290	225	365	232	264	264	300
		157	153	242	168	215	215	300
	(108, 96, 8, 12, 8, 2)	296	225	373	224	264	264	300
		160	153	245	160	215	215	300
V	(147, 133, 9, 15, 8, 2)	409	296	512	305	353	353	412
		216	201	327	218	283	283	412

Table 6 presents the success probabilities of the SDA, NSDA, and NSDA+ algorithms against MAYO_L.

Table 7 compares the public key and signature sizes of MAYO_L with those of the original MAYO scheme.

Table 6: Success probability of SDA, NSDA, and NSDA+

Level	Parameters	SDA	NSDA	NSDA+
I	(75, 63, 9, 7, 8, 2)	2^{-153}	2^{-153}	2^{-153}
	(76, 64, 8, 8, 8, 2)	2^{-156}	2^{-156}	2^{-156}
III	(110, 93, 11, 9, 8, 2)	2^{-294}	2^{-294}	2^{-294}
	(110, 96, 10, 10, 8, 2)	2^{-246}	2^{-246}	2^{-246}
	(108, 96, 8, 12, 8, 2)	2^{-252}	2^{-252}	2^{-252}
V	(147, 133, 9, 15, 8, 2)	2^{-357}	2^{-357}	2^{-357}

Table 7: Key and signature sizes for MAYO and MAYO_L across security levels. All sizes are reported in bytes (B) or kilobytes (KB).

Level	Scheme	Parameters	pk (B)	sig (B)	esk (KB)	epk (KB)
I	MAYO	$(n, m, o, k, q) = (86, 78, 8, 10, 16)$	1420	454	142	143
	MAYO _L	$(n, m, o, k, q, r) = (75, 63, 9, 7, 8, 2)$	1080	418	65	66
	MAYO _L	$(n, m, o, k, q, r) = (76, 64, 8, 8, 8, 2)$	880	480	68	69
III	MAYO	$(n, m, o, k, q) = (118, 108, 10, 11, 16)$	2986	681	368	371
	MAYO _L	$(n, m, o, k, q, r) = (110, 93, 11, 9, 8, 2)$	2318	767	207	208
	MAYO _L	$(n, m, o, k, q, r) = (110, 96, 10, 10, 8, 2)$	1996	849	214	215
	MAYO _L	$(n, m, o, k, q, r) = (108, 96, 8, 12, 8, 2)$	1312	996	206	207
V	MAYO	$(n, m, o, k, q) = (154, 142, 12, 12, 16)$	5554	964	823	828
	MAYO _L	$(n, m, o, k, q, r) = (147, 133, 9, 15, 8, 2)$	2260	1694	529	530

Appendix B: Proof of Theorem 1

In this section, we provide a detailed proof of Theorem 1. The proof is structured into two main lemmas. The first lemma reduces EUF-CMA security to EUF-KOA security by simulating the signing oracle. The second lemma further reduces EUF-KOA security to the hardness of the UOV and MT-WMQ_L⁻ problems.

Reduction from EUF-CMA to EUF-KOA

Lemma 3. *Let $\mathcal{A}$ be an EUF-CMA adversary against the MAYO_L⁻ scheme with the given parameters, making Q_s signing queries and Q_h queries to the random oracle H . Let $\tau = q^{r(m-ko-1)}$ be the empirical failure probability and assume $Q_s\tau < 1$. Then there exists an adversary $\mathcal{B}$ attacking the EUF-KOA security of MAYO_L⁻ running in time $T(\mathcal{A}) + O((Q_h + Q_s)\text{poly}(n, m, k, q, a, r))$ such that*

$$\text{Adv}_{\text{MAYO}_L^-}^{\text{EUF-CMA}}(\mathcal{A}) \leq \text{Adv}_{\text{MAYO}_L^-}^{\text{EUF-KOA}}(\mathcal{B})(1 - Q_s\tau)^{-1} + \frac{(Q_h + Q_s)Q_s}{2^{\text{salt_length}}}.$$

Proof. The adversary $\mathcal{B}$ (attacking EUF-KOA) is constructed as follows. $\mathcal{B}$ receives a public key $pk = (\text{seed}, \{\mathbf{P}_i^{(3)}\}_{i \in [m-a]})$. $\mathcal{B}$ will simulate the environment for $\mathcal{A}$.

Simulation of the random oracle H :

$\mathcal{B}$ maintains a list L of pairs $(X, \mathbf{t})$, initially empty. When $\mathcal{A}$ queries H with input X :

- If $(X, \mathbf{t})$ already exists in L , $\mathcal{B}$ returns $\mathbf{t}$.
- Otherwise, $\mathcal{B}$ samples $\mathbf{t} \leftarrow \mathbb{F}_{q^r}^m$ uniformly at random, adds $(X, \mathbf{t})$ to L , and returns $\mathbf{t}$ to $\mathcal{A}$.

Simulation of the signing oracle:

When $\mathcal{A}$ requests a signature for message M , $\mathcal{B}$ performs the following steps:

1. Choose $\text{salt} \leftarrow \{0, 1\}^{\text{salt_length}}$ uniformly at random.
2. If $(M \parallel \text{salt}, \star)$ already exists in L , $\mathcal{B}$ aborts.
3. Choose $\mathbf{s} = (\mathbf{s}_1, \dots, \mathbf{s}_k) \leftarrow \mathbb{F}_{q^r}^{kn}$ uniformly at random.
4. Compute $\mathbf{t} = \mathcal{P}^*(\mathbf{s})$.
5. Insert $(M \parallel \text{salt}, \mathbf{t})$ into L .
6. Return the signature $(\text{salt}, \mathbf{s})$.

Finally, when $\mathcal{A}$ outputs a forged message-signature pair (M, σ) , $\mathcal{B}$ outputs the same pair.

To analyze the success probability of $\mathcal{B}$, we construct a sequence of games as follows:

- **Game 0:** The EUF-CMA game of $\mathcal{A}$ against MAYO_L⁻. The secret key is generated according to the correct algorithm. The signing oracle uses the real secret key. The random oracle H is a truly random function. By definition,

$$\Pr[\text{Game}_0 = 1] = \text{Adv}_{\text{MAYO}_L^-}^{\text{EUF-CMA}}(\mathcal{A}).$$

- **Game 1:** This game is identical to Game 0, except that if during a signing query the chosen salt collides with a previous random oracle query (i.e., there already exists an entry $(M\|\text{salt}, \mathbf{t})$ in L), the game aborts and outputs 0. The probability of abort for each signing query is at most $(Q_h + Q_s)/2^{\text{salt_length}}$. Hence the total abort probability in the entire game is bounded by $(Q_h + Q_s)Q_s 2^{-\text{salt_length}}$. Thus

$$\Pr[\text{Game}_1 = 1] \geq \Pr[\text{Game}_0 = 1] - (Q_h + Q_s)Q_s 2^{-\text{salt_length}}.$$

- **Game 2:** This game is identical to Game 1, except that if the signing algorithm encounters a restart (due to $\mathcal{P}^*(\mathbf{v} + \mathbf{o})$ not having full rank), the game aborts and outputs 0. From experiments, the probability of restart for one signing attempt is $\tau = q^{r(m-ko-1)}$. Therefore the probability that at least one restart occurs among Q_s signing queries is bounded by $Q_s \tau$. Note that from the adversary's perspective in Game 1, the number of signing attempts is independent: if the signing algorithm encounters a non-full-rank system, it simply restarts internally. Thus the output distribution of the signing algorithm is independent of the number of attempts. We obtain

$$\begin{aligned} \Pr[\text{Game}_2 = 1] &= \Pr[\text{Game}_1 = 1 \wedge \text{no restart}] \\ &= \Pr[\text{Game}_1 = 1] \Pr[\text{no restart}] \\ &\geq \Pr[\text{Game}_1 = 1](1 - Q_s \tau). \end{aligned}$$

- **Game 3:** This game corresponds to the EUF-KOA game executed by $\mathcal{B}$. In this game, signatures are generated by directly sampling $\mathbf{s}$ uniformly at random, and the random oracle is simulated as above.

If Game 2 does not abort, the view of $\mathcal{A}$ in Game 2 and Game 3 is identical. Indeed, if no salt value is reused for the same message, then $\mathcal{B}$ perfectly simulates the random oracle. Moreover, since all linear systems are full rank, signatures are computed as $\mathbf{s} = \mathbf{v} + \mathbf{o}$ where $\mathbf{v}$ is sampled uniformly from $(\mathbb{F}_{q^r}^{n-o} \times \{0\}^o)^k$ and $\mathbf{o}$ is sampled uniformly from $(\mathcal{O} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^r})^k$.

In the original MAYO scheme we have

$$(\mathbb{F}_q^{n-o} \times \{0\}^o) + \mathcal{O} = \mathbb{F}_q^n,$$

hence $\mathbf{s}$ is uniformly distributed over $\mathbb{F}_q^{kn}$.

After lifting we obtain

$$(\mathbb{F}_{q^r}^{n-o} \times \{0\}^o) + (\mathcal{O} \otimes_{\mathbb{F}_q} \mathbb{F}_{q^r}) = \mathbb{F}_{q^r}^n,$$

which implies that the signatures in Game 2 are uniformly distributed over $\mathbb{F}_{q^r}^{kn}$.

Therefore the success probability in Game 2 cannot exceed that in Game 3. Hence

$$\Pr[\text{Game}_2 = 1] < \Pr[\text{Game}_3 = 1] = \text{Adv}_{\text{MAYO}_L^-}^{\text{EUF-KOA}}(\mathcal{B}).$$

Combining the inequalities we obtain

$$\text{Adv}_{\text{MAYO}_L^-}^{\text{EUF-CMA}}(\mathcal{A}) \leq \text{Adv}_{\text{MAYO}_L^-}^{\text{EUF-KOA}}(\mathcal{B})(1 - Q_s \tau)^{-1} + (Q_h + Q_s)Q_s 2^{-\text{salt_length}}.$$

□

Reduction from KOA to UOV + MT-WMQ_L⁻

Lemma 4. *Let $\mathcal{A}$ be an adversary attacking EUF-KOA security of the MAYO_L⁻ scheme. Then there exist adversaries $\mathcal{C}$ and $\mathcal{D}$ running in time $T(\mathcal{A}) + O((1 + Q_h)\text{poly}(n, m, k, q, a, r))$ such that:*

$$\text{Adv}_{\text{MAYO}_L^-}^{\text{EUF-KOA}}(\mathcal{A}) \leq \text{Adv}_{n,m-a,o,q}^{\text{UOV}}(\mathcal{C}) + \text{Adv}_{n,m,k,q,a,r}^{\text{MT-WMQ}_L^-}(\mathcal{D}) + q^{-rm}.$$

Proof. We construct a sequence of games.

- **Game 0:** The real EUF-KOA game. The public key $\mathcal{P}$ is generated from a random oil space $\mathcal{O}$, i.e., $\mathcal{P} \in \text{MQ}_{n,m-a,q}(\mathbf{O})$ with $\mathbf{O} \leftarrow \mathbb{F}_q^{(n-o) \times o}$. The random oracle H is a truly random function. We have

$$\Pr[\text{Game}_0 = 1] = \text{Adv}_{\text{MAYO}_L^-}^{\text{EUF-KOA}}(\mathcal{A}).$$

- **Game 1:** This game is identical to Game 0, except that instead of generating $\mathcal{P}$ from an oil space, $\mathcal{P}$ is chosen uniformly at random from $\text{MQ}_{n,m-a,q}$. Everything else remains unchanged.

We construct an adversary $\mathcal{C}$ against the UOV assumption as follows. When $\mathcal{C}$ is given a multivariate quadratic map $\mathcal{P}$, it computes the matrix representation $\{\mathbf{P}_i^{(1)}, \mathbf{P}_i^{(2)}, \mathbf{P}_i^{(3)}\}_{i \in [m-a]}$ of $\mathcal{P}$. Then $\mathcal{C}$ chooses a random seed and runs $\mathcal{A}$ with input $pk = (\text{seed}, \{\mathbf{P}_i^{(3)}\}_{i \in [m-a]})$, while honestly simulating a random oracle and an Expand oracle. The Expand oracle outputs $\mathbf{P}_i^{(1)}$ on input $\text{seed}||P1||i$, and outputs $\mathbf{P}_i^{(2)}$ on input $\text{seed}||P2||i$. For other inputs, it outputs random matrices. We design $\mathcal{C}$ in such a way that if $\mathcal{C}$ is given $\mathcal{P}$ as a $\text{UOV}(n, m-a, o, q)$ map, then $\mathcal{C}$ exactly corresponds to Game 0, and if $\mathcal{C}$ is given a random map $\mathcal{P}$, then $\mathcal{C}$ corresponds to Game 1. Therefore we have

$$\text{Adv}_{n,m-a,o,q}^{\text{UOV}}(\mathcal{C}) = |\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_1 = 1]|.$$

Construction of adversary $\mathcal{D}$ for the MT-WMQ_L⁻ problem.

Adversary $\mathcal{D}$ receives an instance of the MT-WMQ_L⁻ problem: a random $\mathcal{P}$ from $\text{MQ}_{n,m-a,q}$ and an oracle $\mathcal{O}_t$ that allows access to random target values $\mathbf{t}_i$. $\mathcal{D}$ simulates Game 1 for $\mathcal{A}$ as follows:

1. $\mathcal{D}$ runs $\mathcal{A}$ with public key $\mathcal{P}$.
2. $\mathcal{D}$ simulates the random oracle H using its oracle $\mathcal{O}_t$. Specifically, $\mathcal{D}$ maintains a counter j for distinct queries. When $\mathcal{A}$ makes the j -th query with input X , $\mathcal{D}$ calls the oracle $\mathcal{O}_t$ to obtain $\mathbf{t}_j$ and returns $\mathbf{t}_j$ to $\mathcal{A}$.
3. When $\mathcal{A}$ terminates and outputs a forged message-signature pair (M, σ) with $\sigma = (\text{salt}, \mathbf{s})$, $\mathcal{D}$ checks the validity of the signature by recomputing $\mathcal{P}^*(\mathbf{s})$ and comparing it with $H(M||\text{salt})$.
4. If the signature is valid, then $H(M||\text{salt})$ must be one of the previously returned $\mathbf{t}_j$. Suppose it is $\mathbf{t}_I$. $\mathcal{D}$ outputs $(I, \mathbf{s})$ and wins.
5. If the signature is invalid, $\mathcal{D}$ does nothing further.

In this simulation, the view of $\mathcal{A}$ is identical to that in Game 1, since the values $\mathbf{t}_j$ from the oracle $\mathcal{O}_t$ are random and independent. Therefore $\mathcal{A}$ outputs a valid forgery with probability $\Pr[\text{Game}_1 = 1]$. However, $\mathcal{A}$ may succeed without querying the random oracle on input $M||\text{salt}$.

This probability is at most q^{-rm} , since there are q^{rm} possible values for $\mathbf{t}$. If a forgery occurs and $\mathcal{A}$ has queried the oracle on $M||\text{salt}$ (the I -th query), then $\mathcal{D}$ succeeds. Therefore

$$\Pr[\text{Game}_1 = 1] \leq \Pr[\mathcal{D} \text{ succeeds}] + q^{-rm} = \text{Adv}_{n,m,k,q,a,r}^{\text{MT-WMQ}_L^-}(\mathcal{D}) + q^{-rm}.$$

Combining the results, we obtain

$$\Pr[\text{Game}_0 = 1] \leq \Pr[\text{Game}_1 = 1] + \text{Adv}_{n,m-a,o,q}^{\text{UOV}}(\mathcal{C}) \leq \text{Adv}_{n,m,k,q,a,r}^{\text{MT-WMQ}_L^-}(\mathcal{D}) + \text{Adv}_{n,m-a,o,q}^{\text{UOV}}(\mathcal{C}) + q^{-rm}.$$

□

Appendix C: EUF-CMA in QROM

In this section, we present a security proof in the QROM based on [46]. Since this proof relies on the assumption of Weak Preimage-Sampleable Functions (WPSF), it is not as tight as the previous proof in the ROM.

Definition 3 (Non-invertibility (INV)). *Let $\mathsf{T} = (\text{Gen}, \mathsf{F}, \mathsf{I})$ be a trapdoor function. Consider the following inversion game between a challenger and an adversary $\mathcal{B}$.*

1. **Setup:** $(\mathsf{F}, \mathsf{I}) \leftarrow \text{Gen}(1^\lambda)$.
2. **Challenge:** Sample $\mathbf{y} \xleftarrow{\$} \mathcal{Y}$, where $\mathcal{Y}$ is the image of F .
3. **Response:** The adversary $\mathcal{B}$ is given $(\mathsf{F}, \mathbf{y})$ and outputs $\mathbf{x} \in \mathcal{X}$.
4. **Outcome:** $\mathcal{B}$ wins if $\mathsf{F}(\mathbf{x}) = \mathbf{y}$.

The advantage of $\mathcal{B}$ in the inversion game is defined as

$$\text{Adv}_{\mathsf{T}}^{\text{INV}}(\mathcal{B}) = \Pr[\mathcal{B} \text{ wins}].$$

The trapdoor function T is said to be non-invertible if for every probabilistic polynomial-time adversary $\mathcal{B}$, the advantage $\text{Adv}_{\mathsf{T}}^{\text{INV}}(\mathcal{B})$ is negligible in the security parameter λ .

Theorem 2. *For any quantum EUF-CMA adversary $\mathcal{A}$ of MAYO_L^- issuing at most q_{sign} classical queries to the signing oracle and q_{qro} (quantum) random oracle queries to the hash function (with $q_{\text{sign}}\tau < 1$), there exists an INV adversary $\mathcal{B}_{\text{inv}}$ of $\mathsf{T}_{\text{MAYO}_L^-}$ such that*

$$\text{Adv}_{[\mathsf{T}_{\text{mayo}_L^-}, H]}^{\text{EUF-CMA}}(\mathcal{A}) \leq \frac{(2q_{\text{qro}} + 1)^2}{1 - q_{\text{sign}}\tau} \text{Adv}_{\mathsf{T}_{\text{mayo}_L^-}}^{\text{INV}}(\mathcal{B}_{\text{inv}}) + \frac{3}{2}q_{\text{sign}}\sqrt{\frac{q_{\text{sign}} + q_{\text{qro}} + 1}{2\text{salt_length}}},$$

where the running time of $\mathcal{B}_{\text{inv}}$ is about that of $\mathcal{A}$.

Proof idea. We follow the proof methodology of [46, Theorem 1 and Proposition 6] for hash-and-sign with retry in the QROM, adapted to the specific structure of MAYO_L^- . The proof consists of two main steps.

We start from the original EUF-CMA game G_0 . Define G'_1 as the same game but with an abort if any signing query requires a retry (if the linear system $\mathcal{P}^*(\mathbf{v} + \mathbf{o}) = \mathbf{t}$ does not have full rank).

From the empirical failure probability $\tau = q^{r(m-ko-1)}$, the probability that no abort occurs in q_{sign} queries is at least $1 - q_{\text{sign}}\tau$. Hence

$$\Pr[G_0 \Rightarrow 1] \leq \frac{1}{1 - q_{\text{sign}}\tau} \Pr[G'_1 \Rightarrow 1].$$

In G'_1 , all signatures are generated without retry, so the distribution of the signature $\mathbf{s} = \mathbf{v} + \mathbf{o}$ is uniform over $\mathbb{F}_{q^r}^{kn}$ (by the property $(\mathbb{F}_{q^r}^{n-o} \times \{0\}^o) + (\mathcal{O} \otimes \mathbb{F}_{q^r}) = \mathbb{F}_{q^r}^n$).

Now we work entirely within G'_1 . We apply the adaptive reprogramming technique [47] to simulate the signing oracle without the trapdoor. This introduces the term $\frac{3}{2}q_{\text{sign}}\sqrt{\frac{q_{\text{sign}}+q_{\text{qro}}+1}{2^{\text{salt_length}}}}$ (from the reprogramming lemma [46, Lemma 1]). Then, using the measure-and-reprogram technique [48], we embed an INV challenge $\mathbf{y}$ into one of the random oracle queries. This yields a factor $(2q_{\text{qro}}+1)^2$ and a reduction from G'_1 to the INV problem of the underlying trapdoor function (which is the UOV problem with parameters $(n, m-a, o, q)$). Consequently,

$$\Pr[G'_1 \Rightarrow 1] \leq (2q_{\text{qro}}+1)^2 \text{Adv}^{\text{INV}}(\mathcal{B}_{\text{inv}}).$$

Putting everything together gives the final bound:

$$\text{Adv}_{\text{MAYO}_L^-}^{\text{EUF-CMA}}(\mathcal{A}) \leq \frac{(2q_{\text{qro}}+1)^2}{1 - q_{\text{sign}}\tau} \text{Adv}^{\text{INV}}(\mathcal{B}_{\text{inv}}) + \frac{3}{2}q_{\text{sign}}\sqrt{\frac{q_{\text{sign}}+q_{\text{qro}}+1}{2^{\text{salt_length}}}}.$$

This sketch follows the structure used in [46] and has been applied to MAYO [15]. $\square$